Lab Project Report

Ethical Hacking Lab

IV-Year, Semester-I

2026-2027

Student Name : E Spoorthi

Roll No : 23B81A6253

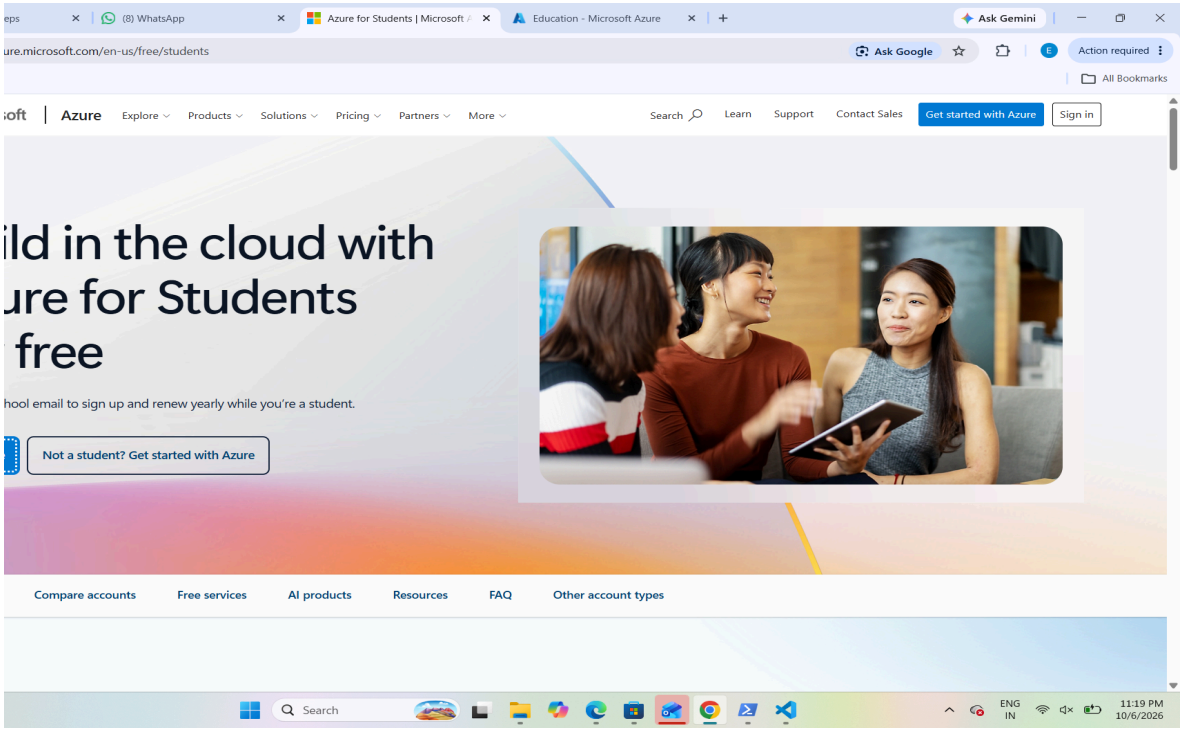
Section : CSC-A

Microsoft Sentinel–Based SOC Attack Simulation and Investigation

Using the Microsoft Sentinel environment configured with an Azure Windows Virtual Machine, Log Analytics Workspace, Windows Security Events via Azure Monitor Agent (AMA), and a Data Collection Rule, simulate controlled security activities on the isolated Windows VM and investigate the resulting security telemetry in Microsoft Sentinel.Generate and analyze events related to failed/successful authentication, password-spray activity, local account creation, privilege modification, process execution, file creation, DNS/network activity, and other permitted lab activities.Use KQL to correlate the events, construct an incident timeline, identify the affected account and host, develop appropriate detection logic, and document the investigation findings and recommended response action.

Steps to Sign in

1. Open the Azure for Students page: Azure for Students
2. Click Sign in at the top-right corner.
3. Enter your college/school Microsoft account credentials.
4. Complete the verification if Microsoft asks for OTP/MFA.
5. After successful login, you will be redirected to the Azure Portal.
6. Open the Azure Portal: Microsoft Azure Portal
7. Check the account shown in the top-right corner.
8. To verify your student subscription, search for Subscriptions in the Azure Portal.
9. Select Azure for Students and verify that the subscription is Active.

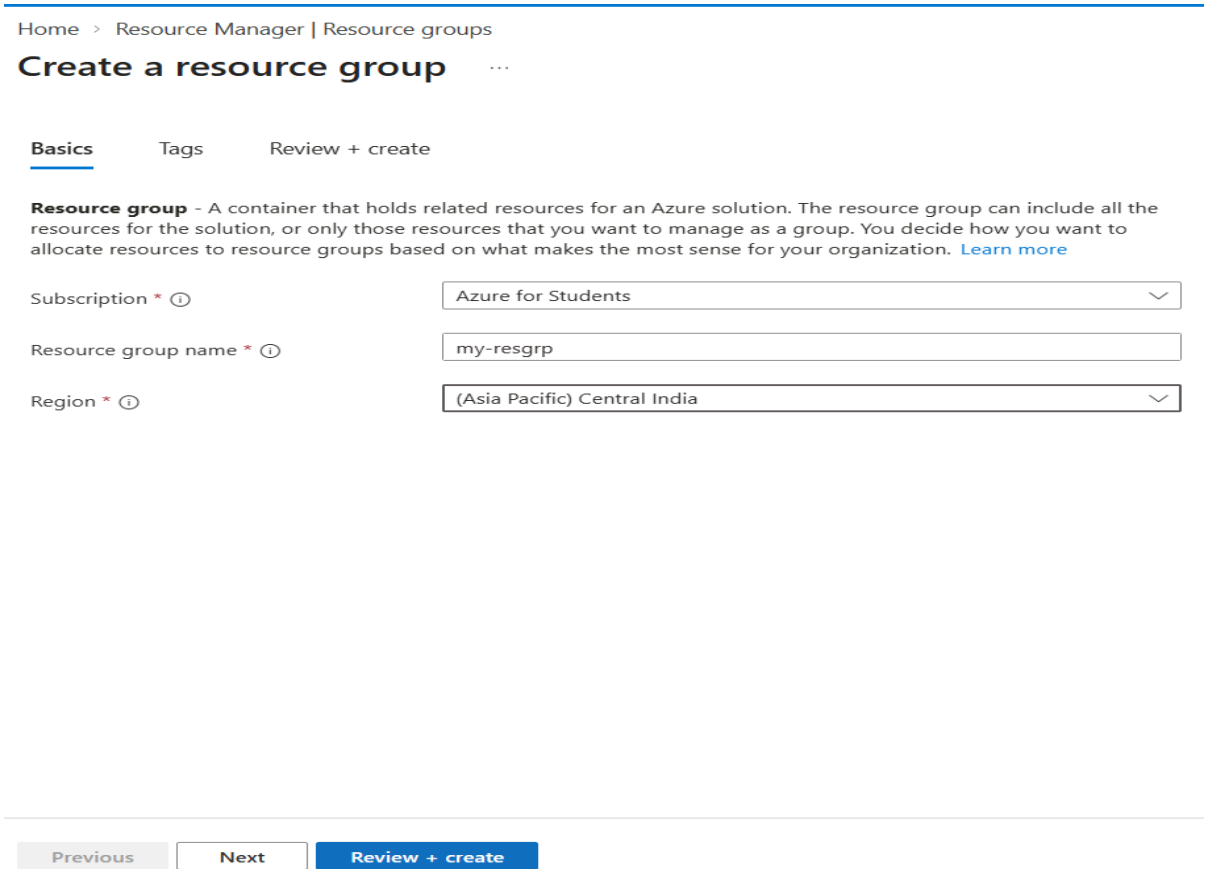


Create a Resource Group

A Resource Group was created to logically organize all resources used in the SOC lab.

1. In the Azure Portal, search for **Resource Groups**.
2. Select **Create**.
3. Select the Azure for Students subscription.
4. Enter a suitable resource group name.
5. Select an appropriate Azure region.
6. Click **Review + create**.
7. After validation is completed, click **Create**.

The Resource Group acts as a container for the virtual machine, monitoring resources, and other components used in the lab.



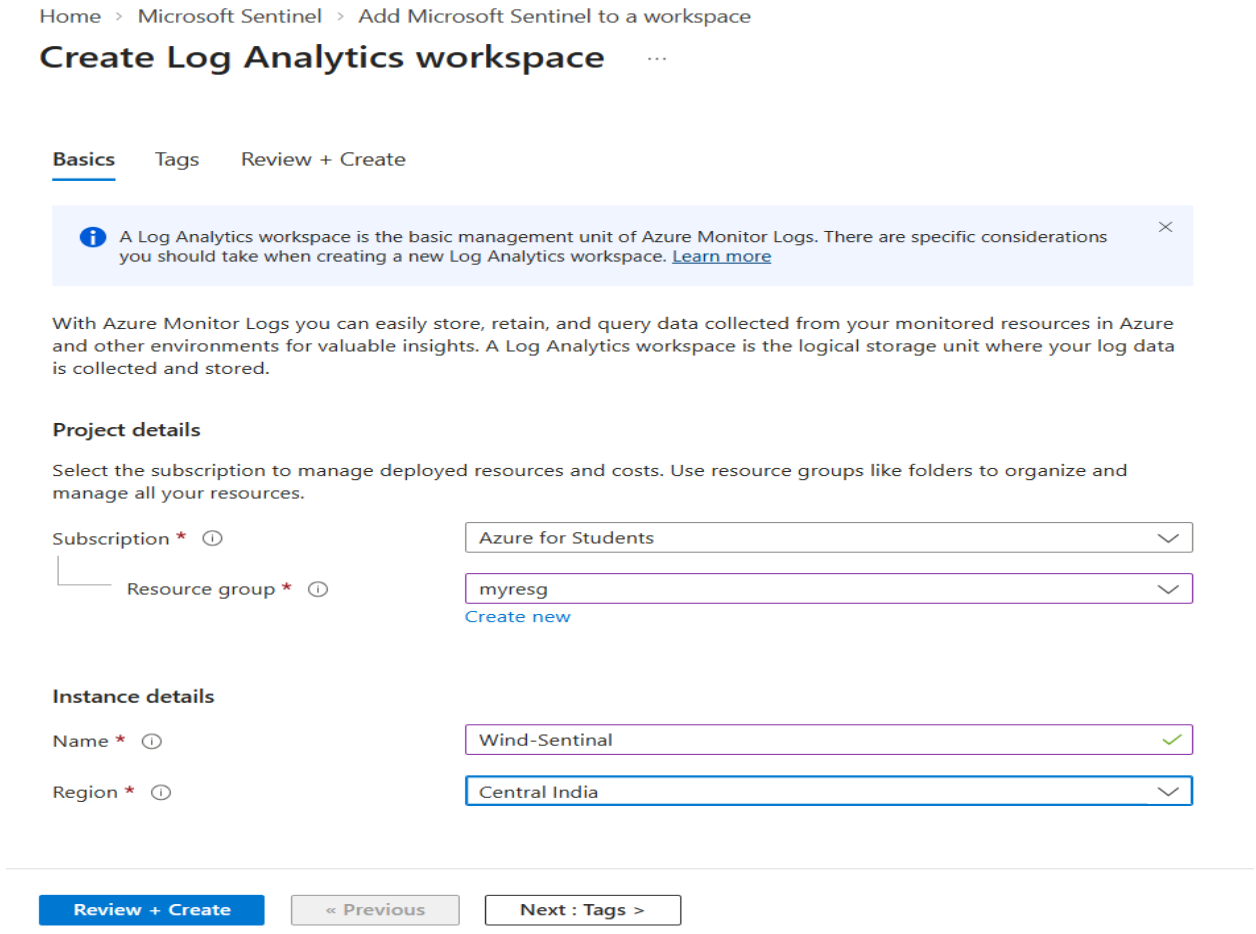
Create Log Analytics Workspace in Microsoft Sentinel

As part of the Microsoft Sentinel setup, a **Log Analytics Workspace** was created to provide the central storage and analysis location for security and system logs.

The workspace was configured with the following details:

- **Subscription:** Azure for Students
- **Resource Group:** `myresg`
- **Workspace Name:** `Wind-Sentinal`
- **Region:** Central India

After entering the required configuration, the workspace was reviewed and created successfully. It was then used as the workspace for the Microsoft Sentinel security monitoring environment.

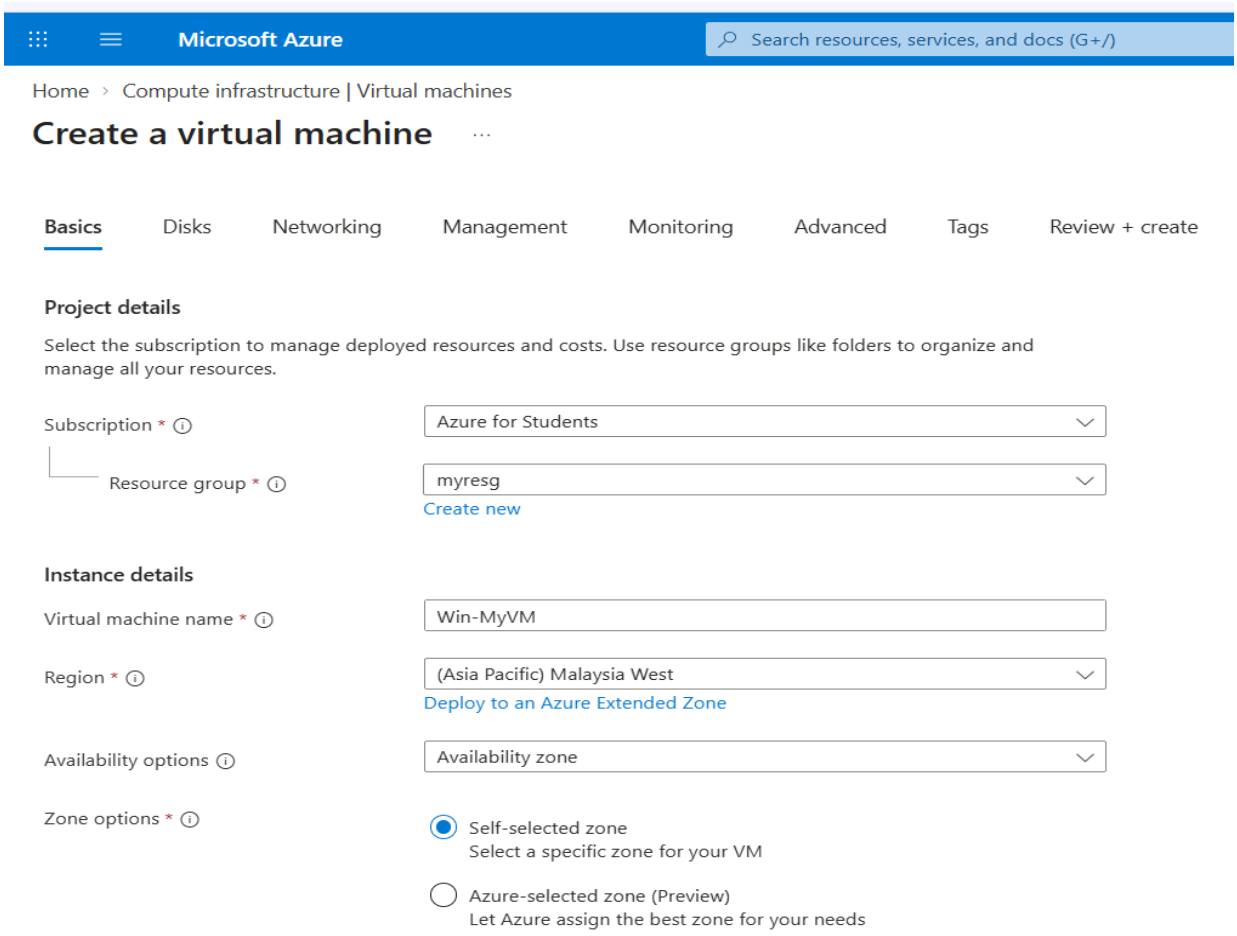


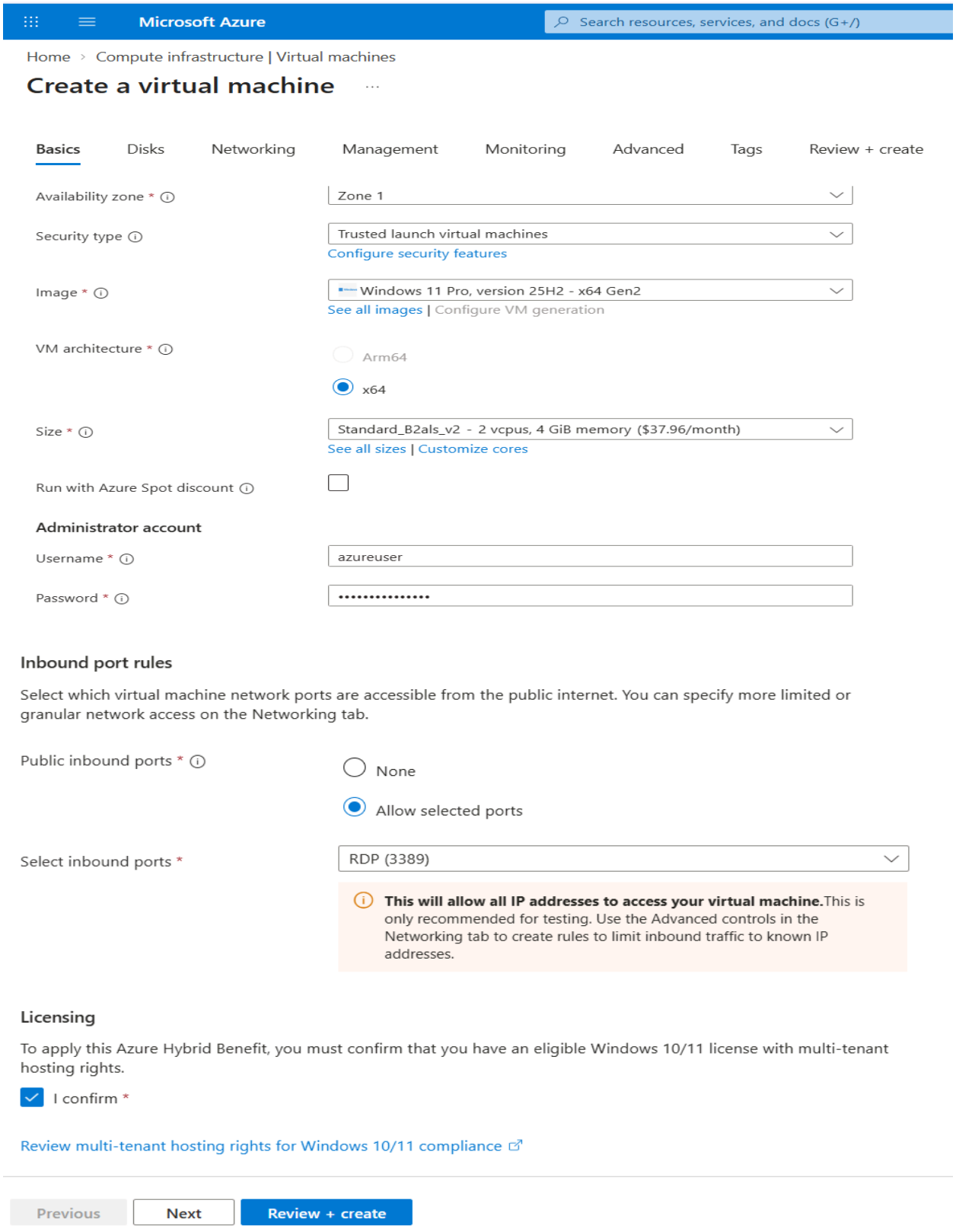
Create the Windows Virtual Machine

A Windows virtual machine was created to act as the monitored endpoint for the SOC lab.

1. Search for **Virtual Machines** in the Azure Portal.
2. Click **Create** → **Azure virtual machine**.
3. Select the previously created Resource Group , Provide a VM name.
5. Select a suitable Windows Server/Windows operating system image.
6. Select an appropriate VM size within the available Azure for Students quota.
7. Configure the administrator username and password.
8. Review the configuration.
9. Click **Create**.
10. Wait until Azure completes the deployment.

The Windows VM becomes the main endpoint on which Windows Security logs, PowerShell activity, and Sysmon telemetry are generated.



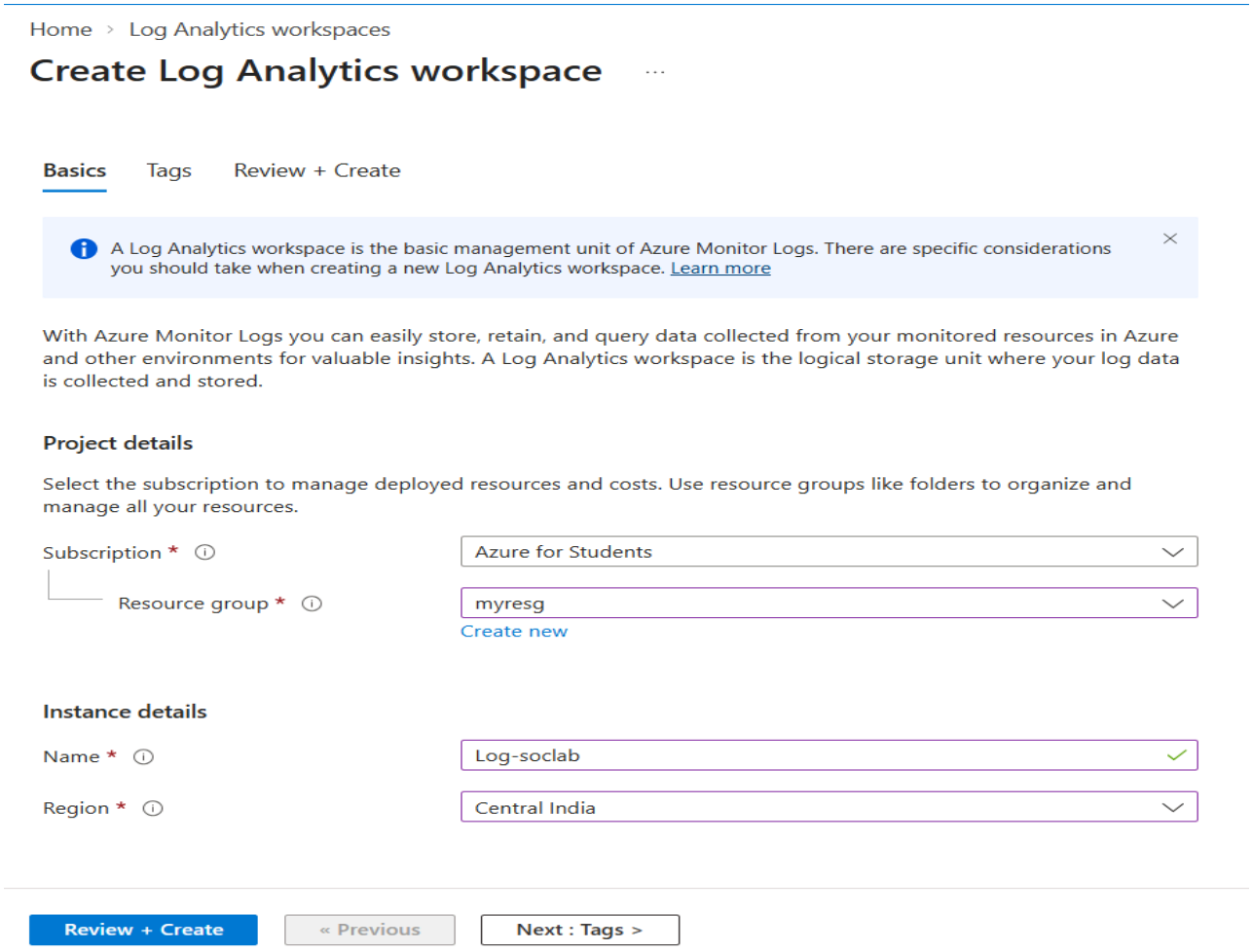

Microsoft Azure
Search resources, services, and docs (G+/)
Home > Compute infrastructure | Virtual machines
Create a virtual machine
Basics
Disks
Networking
Management
Monitoring
Advanced
Tags
Review + create
Availability zone *
Zone 1
Security type
Trusted launch virtual machines
Configure security features
Image *
Windows 11 Pro, version 25H2 - x64 Gen2
See all images | Configure VM generation
VM architecture *
Arm64
x64
Size *
Standard_B2als_v2 - 2 vcpus, 4 GiB memory ($37.96/month)
See all sizes | Customize cores
Run with Azure Spot discount
Administrator account
Username *
azureuser
Password *
Inbound port rules
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.
Public inbound ports *
None
Allow selected ports
Select inbound ports *
RDP (3389)
This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.
Licensing
To apply this Azure Hybrid Benefit, you must confirm that you have an eligible Windows 10/11 license with multi-tenant hosting rights.
I confirm *
Review multi-tenant hosting rights for Windows 10/11 compliance
Previous
Next
Review + create

Create the Log Analytics Workspace

A Log Analytics Workspace was created to collect and store monitoring and security data generated by the Windows VM.

1. In Azure Portal, search for **Log Analytics Workspaces**.
2. Click **Create**.
3. Select the same Azure subscription.
4. Select the previously created Resource Group.
5. Enter a workspace name.
6. Select the appropriate region.
7. Click **Review + create**.
8. Click **Create** after validation.

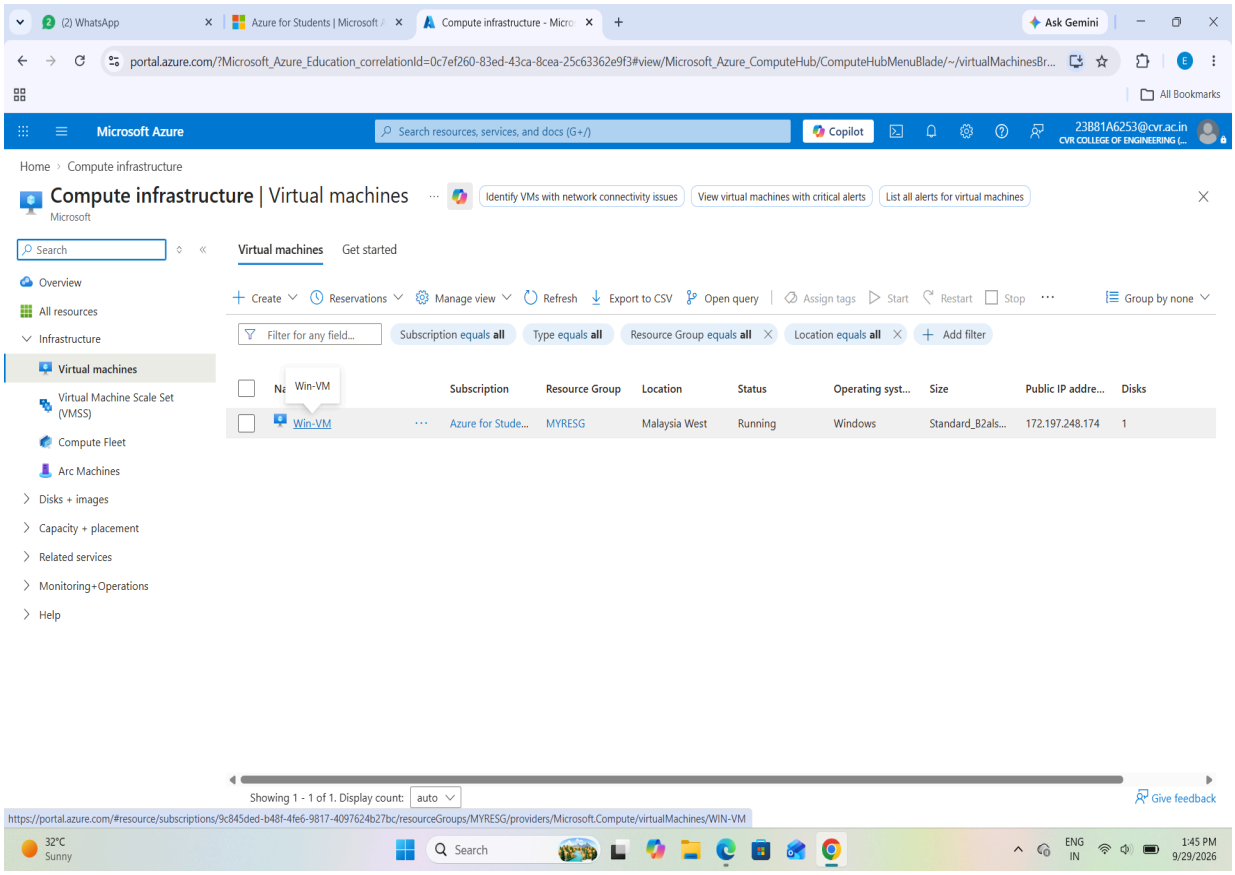
The Log Analytics Workspace acts as the central location where collected telemetry can be queried using Kusto Query Language (KQL).

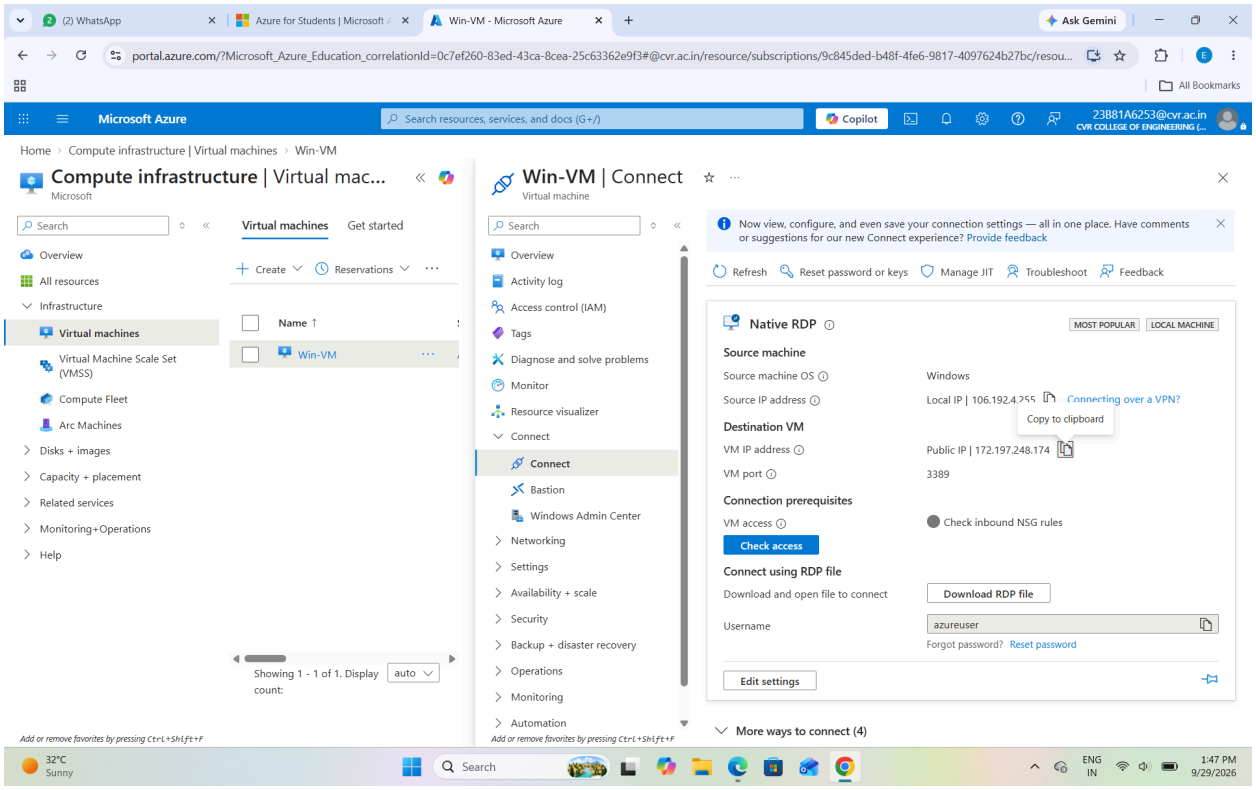


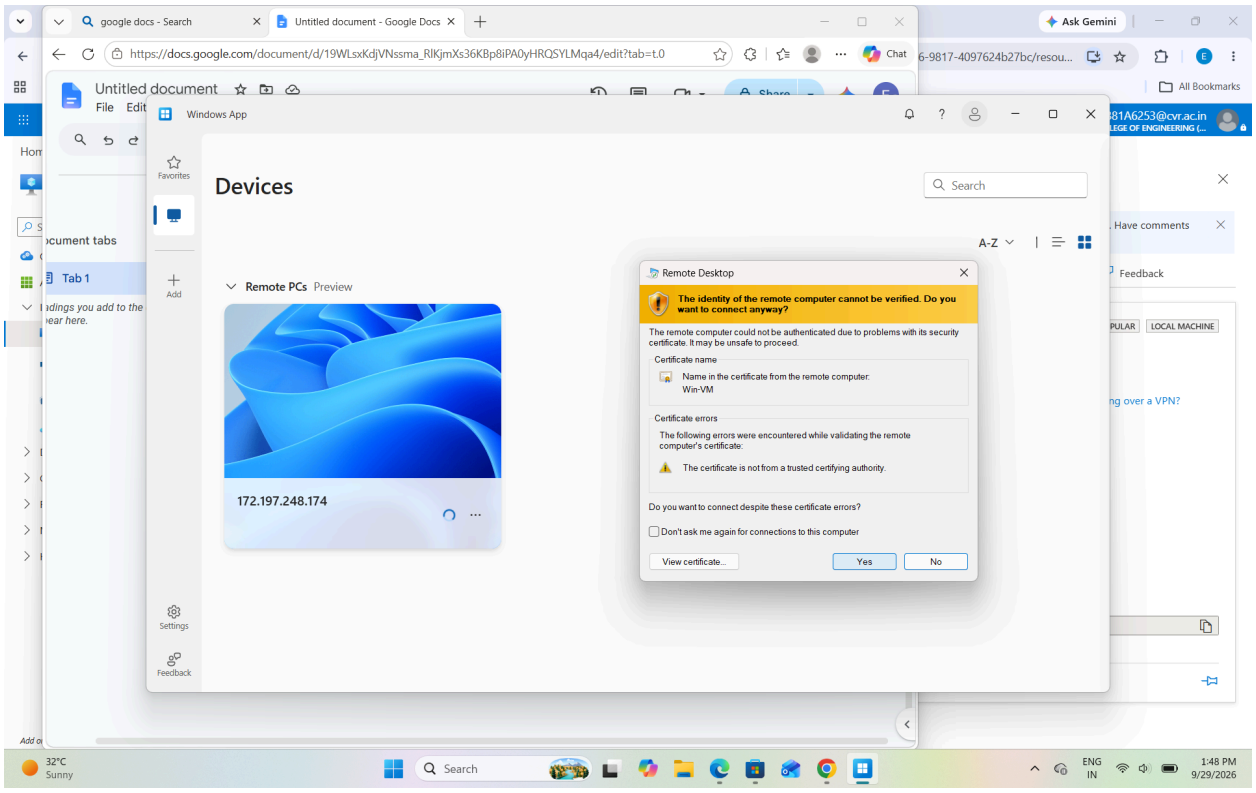
Connect to the Windows VM

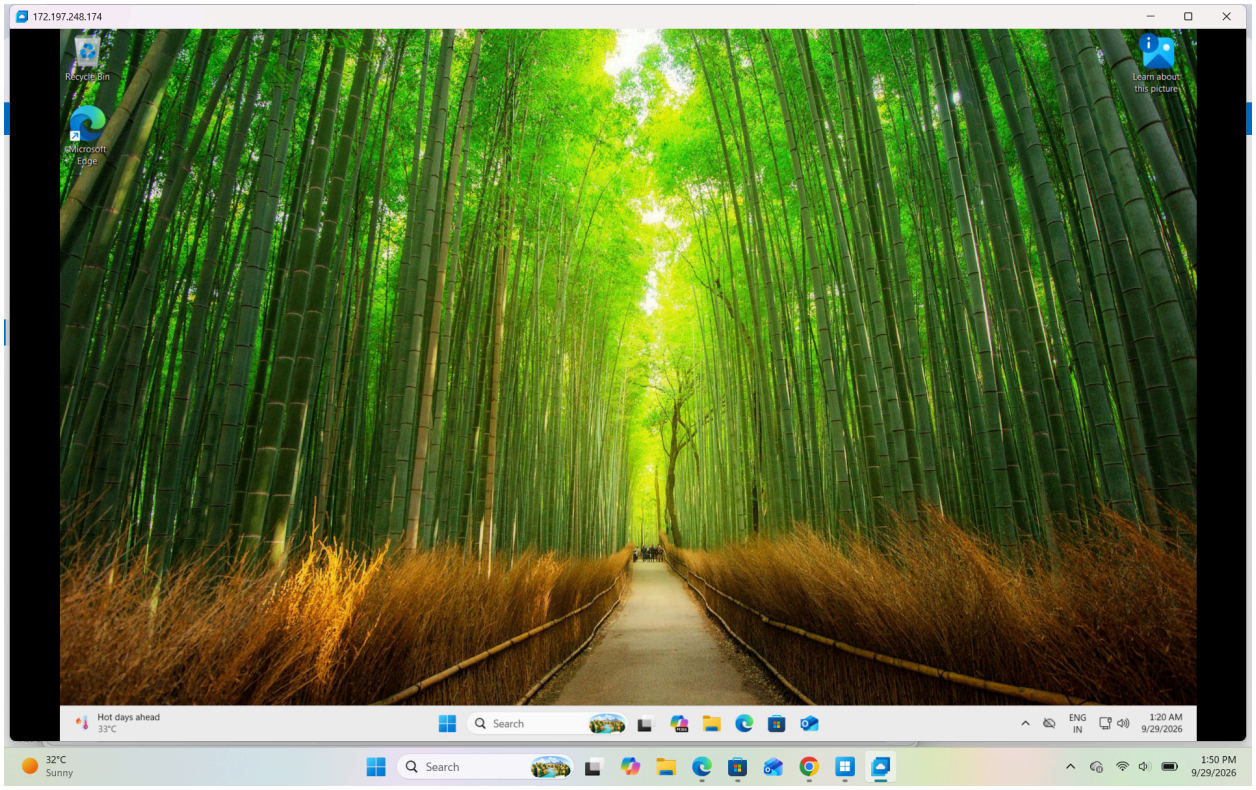
After deployment, the VM was accessed using Remote Desktop Protocol (RDP).

1. Open the created VM in Azure Portal.
2. Click **Connect** → **RDP**.
3. Download the RDP file.
4. Open the RDP file on the local computer.
5. Enter the administrator credentials configured during VM creation.
6. After successful authentication, the Windows desktop of the Azure VM is displayed.




Win-VM - Microsoft Azure
Microsoft Azure
Home > Compute infrastructure | Virtual machines > Win-VM
Compute infrastructure | Virtual mac...
Win-VM | Connect
Virtual machine
Native RDP
Source machine
Destination VM
Connection prerequisites
Check access
Connect using RDP file
Download RDP file
azureuser
Edit settings
More ways to connect (4)


Untitled document - Google Docs
Windows App
Devices
Remote PCs Preview
172.197.248.174
Remote Desktop
The identity of the remote computer cannot be verified. Do you want to connect anyway?
Yes
No



1. Create Administrator Account

Created `soc_admin01` using Command Prompt and added it to the Administrators group.

The account was verified using `net user` and `net localgroup Administrators`

2. Create Standard Users

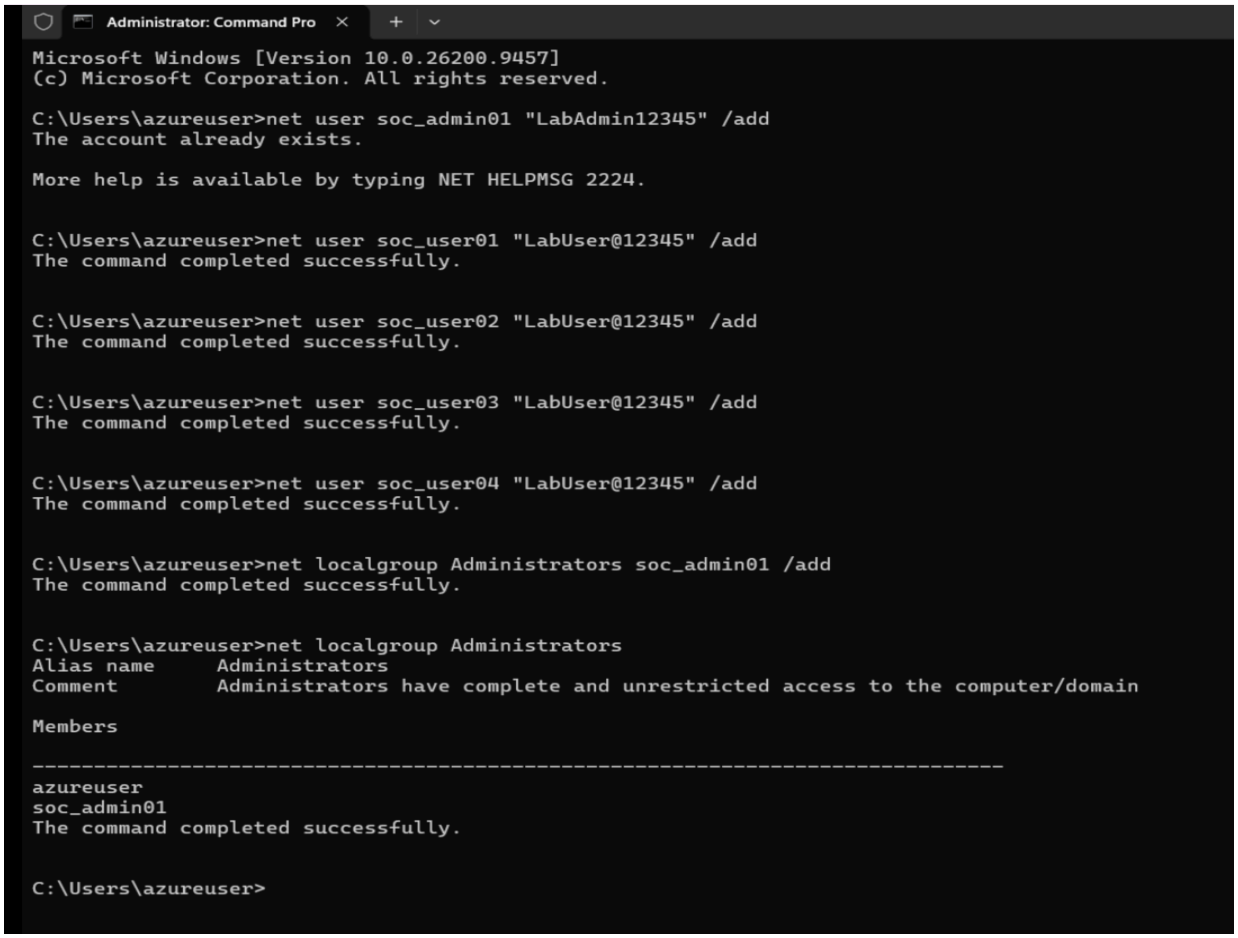
Created four standard accounts:

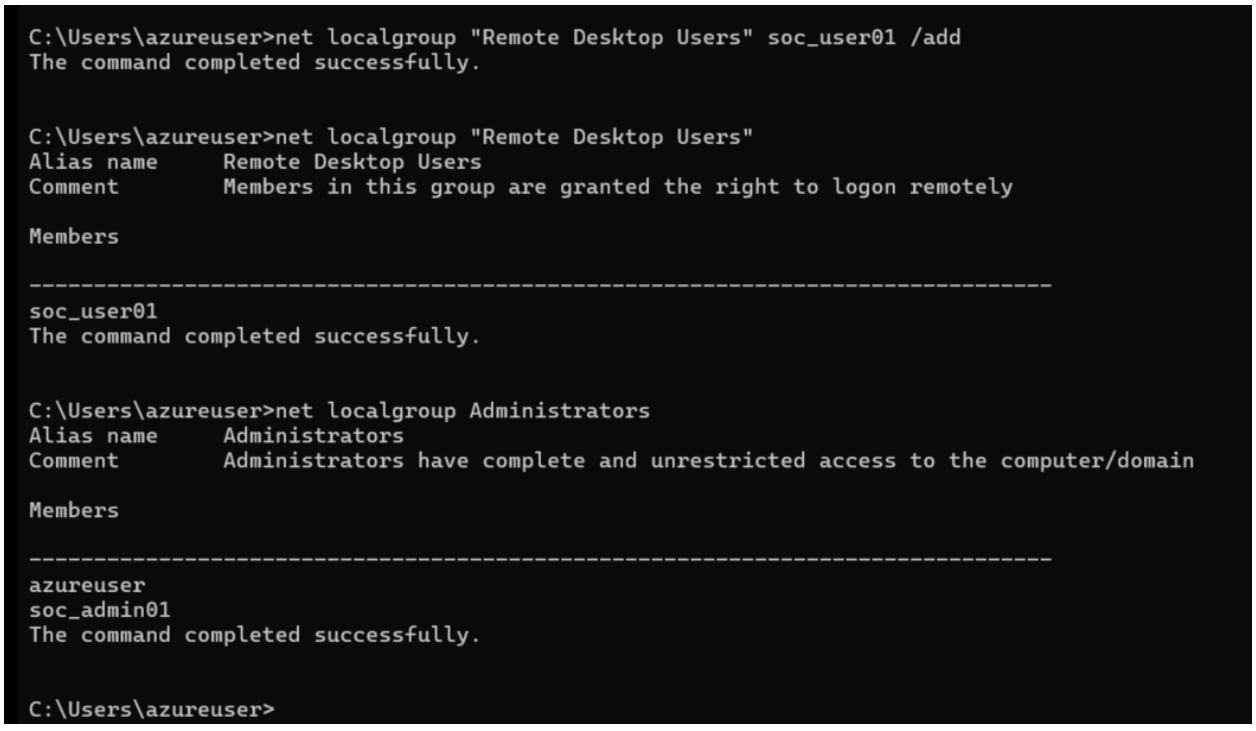
- `soc_user01`
- `soc_user02`
- `soc_user03`
- `soc_user04`

`soc_user01` was added to the **Remote Desktop Users** group so it could be used for RDP while remaining a standard user.

3. Verify User Privileges

Checked the Administrators group to ensure that only `soc_admin01` had administrator privileges. The standard users were kept as normal users for later attack simulations.


Administrator: Command Pro
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\azureuser>net user soc_admin01 "LabAdmin12345" /add
The account already exists.
More help is available by typing NET HELPMSG 2224.
C:\Users\azureuser>net user soc_user01 "LabUser@12345" /add
The command completed successfully.
C:\Users\azureuser>net user soc_user02 "LabUser@12345" /add
The command completed successfully.
C:\Users\azureuser>net user soc_user03 "LabUser@12345" /add
The command completed successfully.
C:\Users\azureuser>net user soc_user04 "LabUser@12345" /add
The command completed successfully.
C:\Users\azureuser>net localgroup Administrators soc_admin01 /add
The command completed successfully.
C:\Users\azureuser>net localgroup Administrators
Alias name Administrators
Comment Administrators have complete and unrestricted access to the computer/domain
Members
azureuser
soc_admin01
The command completed successfully.
C:\Users\azureuser>

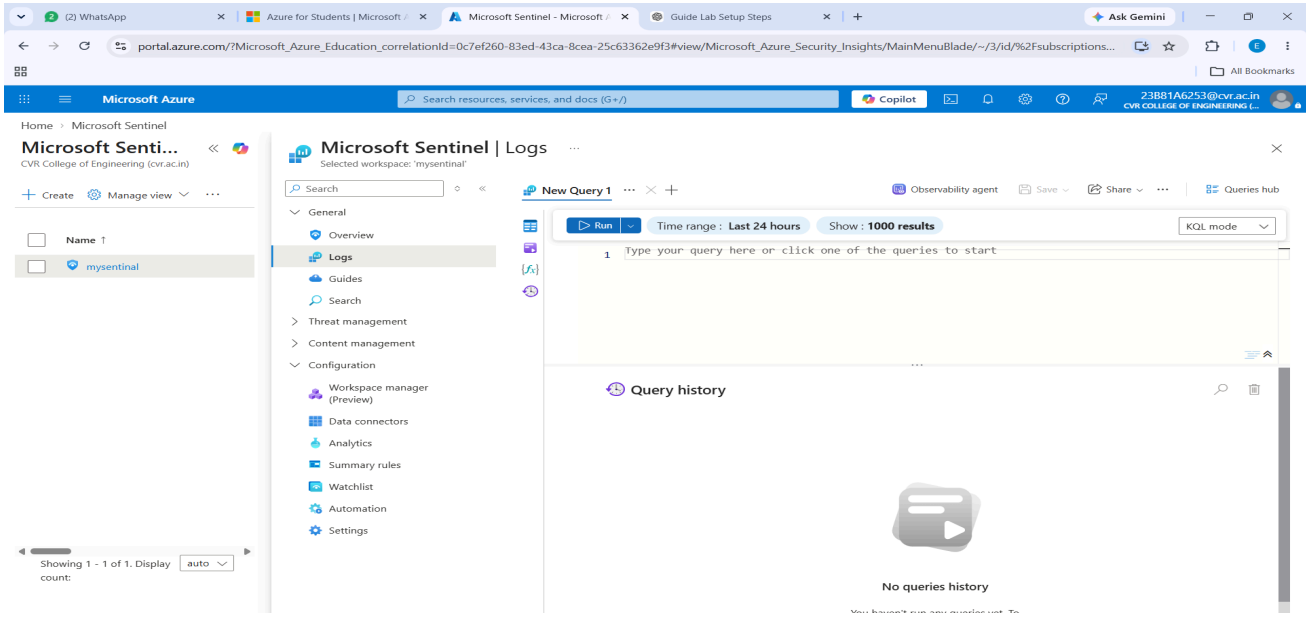

C:\Users\azureuser>net localgroup "Remote Desktop Users" soc_user01 /add
The command completed successfully.
C:\Users\azureuser>net localgroup "Remote Desktop Users"
Alias name Remote Desktop Users
Comment Members in this group are granted the right to logon remotely
Members
soc_user01
The command completed successfully.
C:\Users\azureuser>net localgroup Administrators
Alias name Administrators
Comment Administrators have complete and unrestricted access to the computer/domain
Members
azureuser
soc_admin01
The command completed successfully.
C:\Users\azureuser>

Microsoft Sentinel Logs Workspace

This screenshot shows the **Microsoft Sentinel → Logs** page for the `mysentinel` workspace. The **Logs** section is used to run **KQL (Kusto Query Language) queries** and investigate security data collected from the Windows VM. It is the main interface used throughout the lab to search, analyze, and investigate events generated during the attack simulations.

How to Open Microsoft Sentinel Logs:

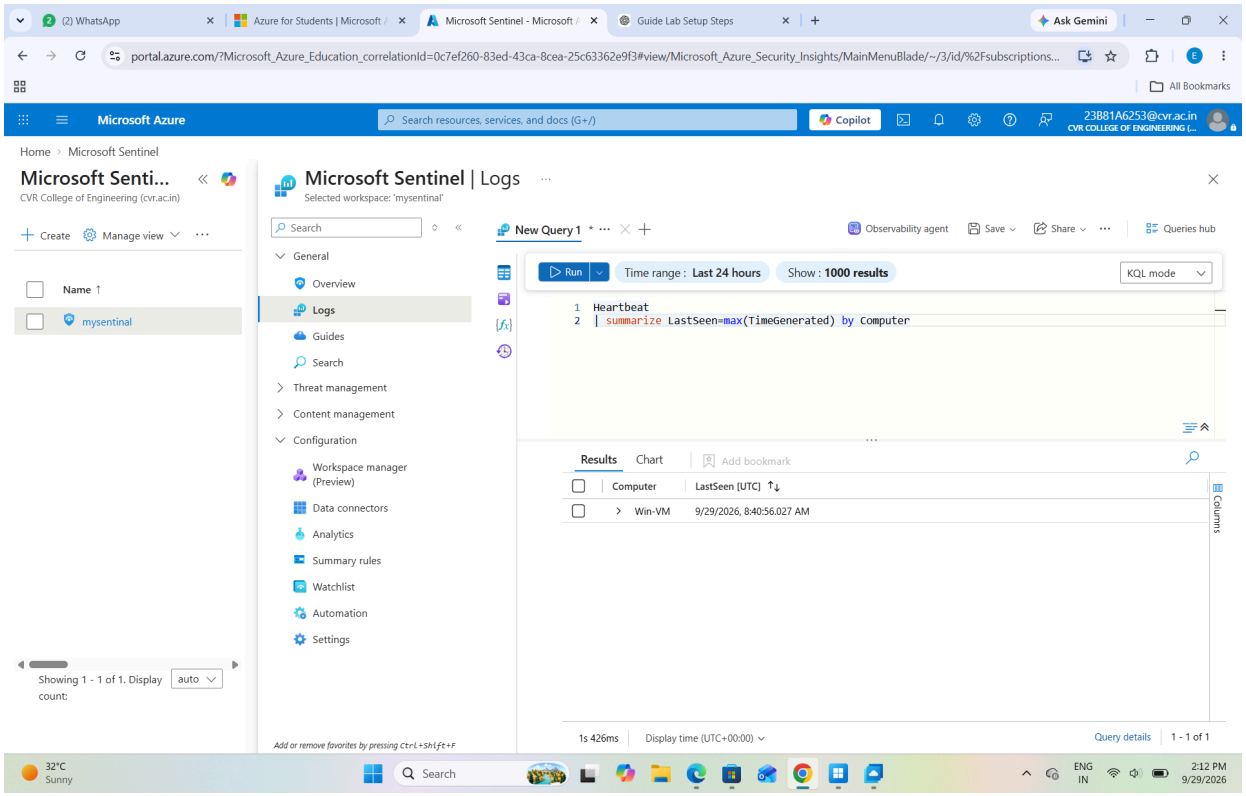
1. Sign in to the **Microsoft Azure Portal**.
2. From the Azure Portal, search for **Microsoft Sentinel** in the search bar.
3. Open **Microsoft Sentinel** from the search results.
4. Select the Sentinel workspace used for the lab, **`mysentinel`**.
5. From the left-side menu, select **Logs** under the **General** section.
6. The **Logs** page will open with the KQL query editor.
7. Enter the required KQL query in the editor and click **Run**.
8. The query results will be displayed in the **Results** section below the editor.



In **Microsoft Sentinel → Logs**, run:

Heartbeat
| summarize LastSeen=max(TimeGenerated) by Computer

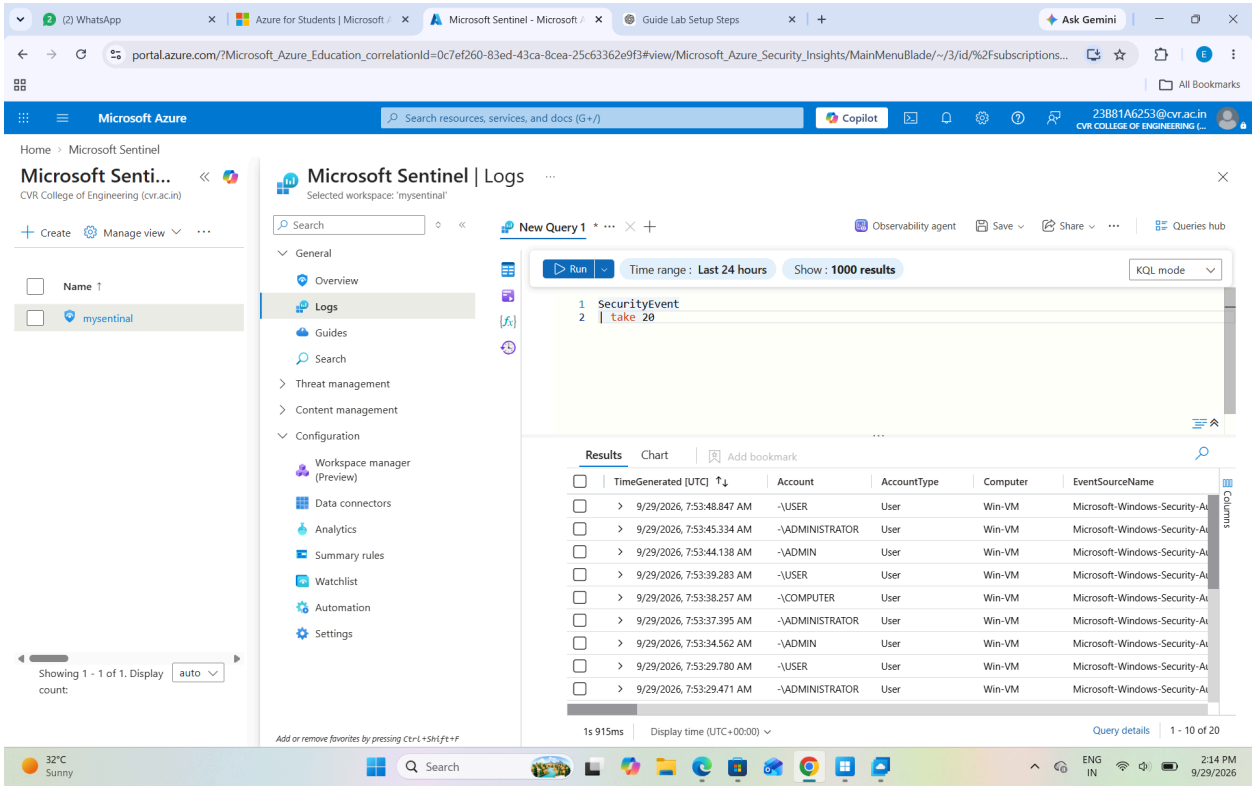
This confirms that the Windows VM is connected and sending data to Sentinel.



Check Security Events
Run:

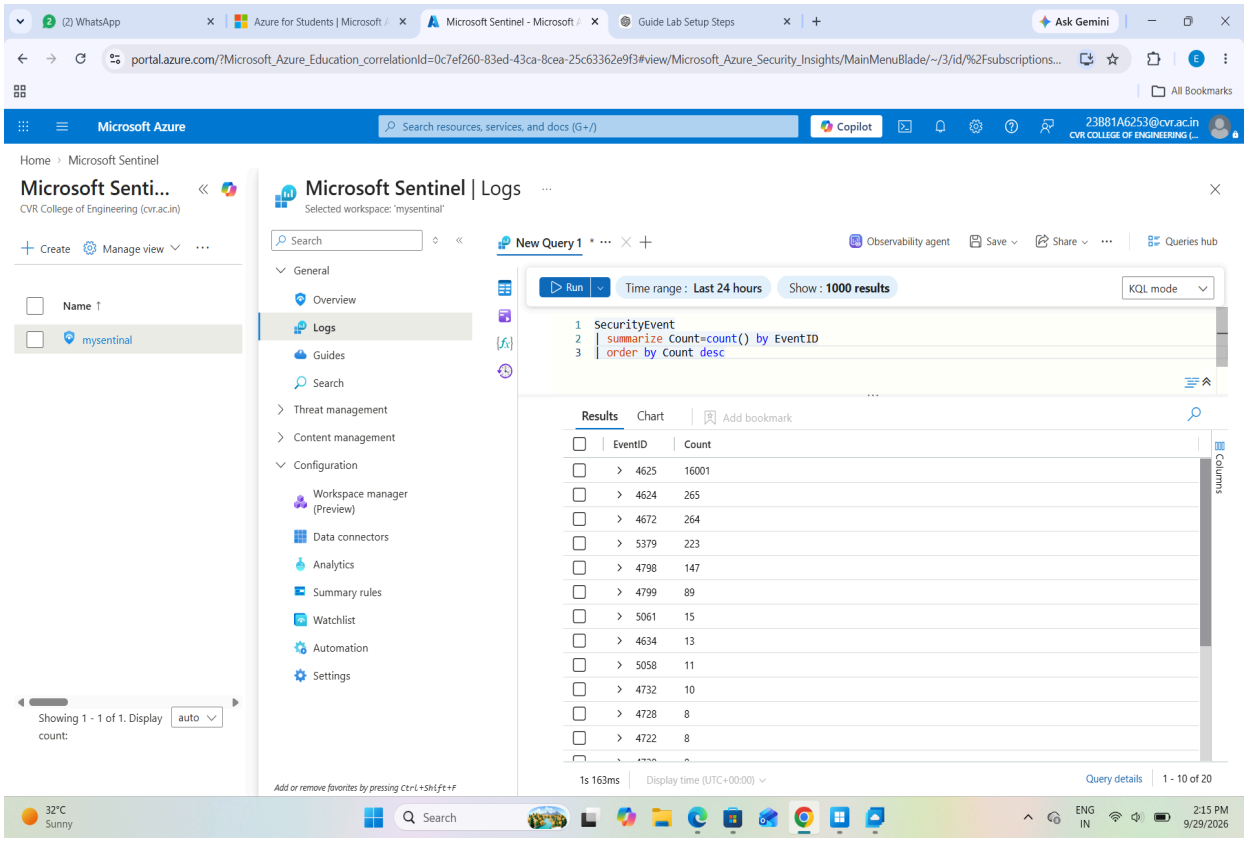
SecurityEvent
| take 20

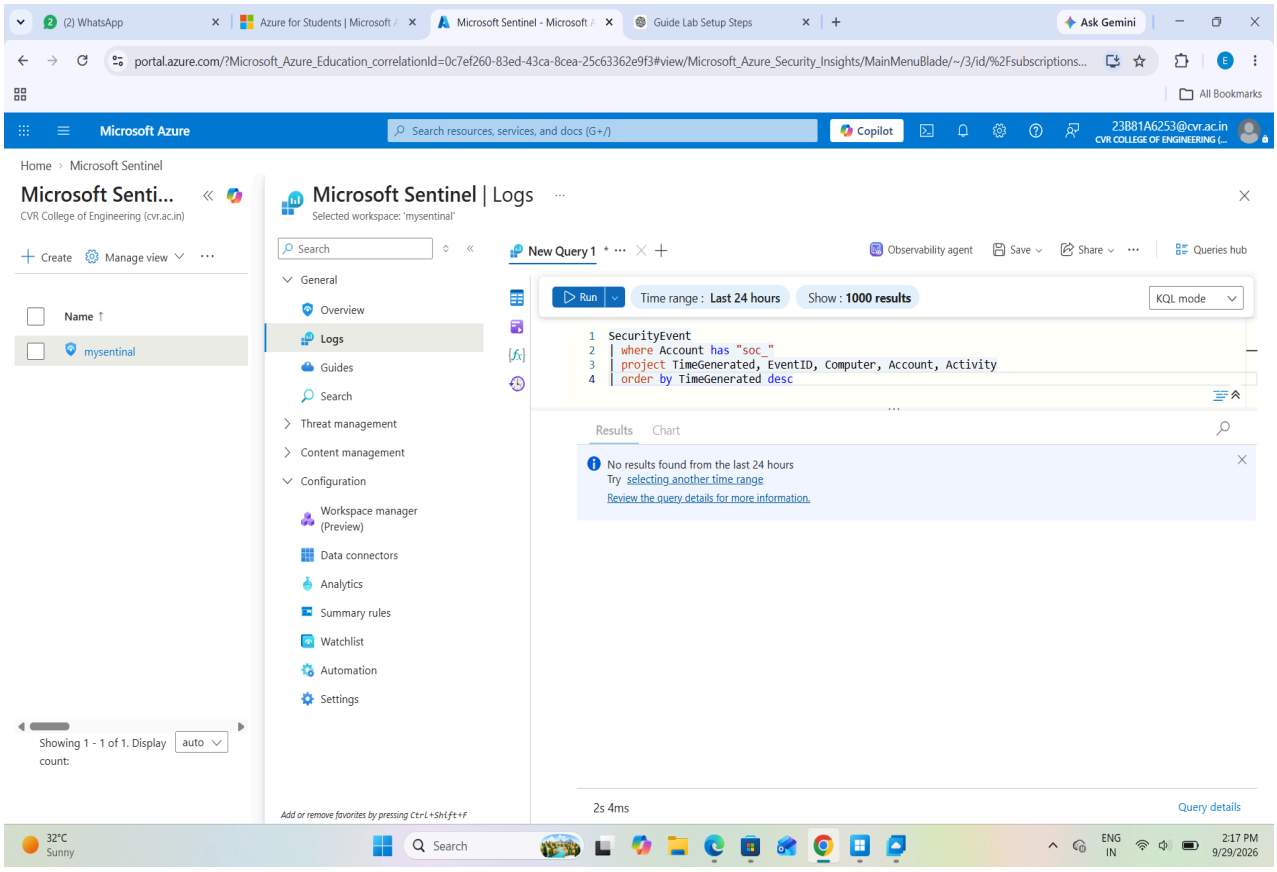
If records are displayed, Windows Security Events are being successfully collected by Sentinel.



To see which event IDs are available:

SecurityEvent
| summarize Count=count() by EventID
| order by Count desc


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home › Microsoft Sentinel
Microsoft Senti...
CVR College of Engineering (cvr.ac.in)
Create
Manage view
Name ↑
mysentinal
Showing 1 - 1 of 1. Display count:
auto
Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
Search
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Add or remove favorites by pressing Ctrl+Shift+F
New Query 1 *
Observability agent
Save
Share
Queries hub
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
1 SecurityEvent
2 | summarize Count=count() by EventID
3 | order by Count desc
Results
Chart
Add bookmark
EventID
Count
4625 16001
4624 265
4672 264
5379 223
4798 147
4799 89
5061 15
4634 13
5058 11
4732 10
4728 8
4722 8
Columns
1s 163ms
Display time (UTC+00:00)
Query details
1 - 10 of 20
32°C
Sunny
Search
ENG
IN
2:15 PM
9/29/2026


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home › Microsoft Sentinel
Microsoft Senti...
CVR College of Engineering (cvr.ac.in)
Create
Manage view
Name ↑
mysentinal
Showing 1 - 1 of 1. Display count:
auto
Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
Search
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Add or remove favorites by pressing Ctrl+Shift+F
New Query 1 *
Observability agent
Save
Share
Queries hub
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
1 SecurityEvent
2 | where Account has "soc_"
3 | project TimeGenerated, EventID, Computer, Account, Activity
4 | order by TimeGenerated desc
Results
Chart
No results found from the last 24 hours
Try selecting another time range
Review the query details for more information.
2s 4ms
Query details
32°C
Sunny
Search
ENG
IN
2:17 PM
9/29/2026

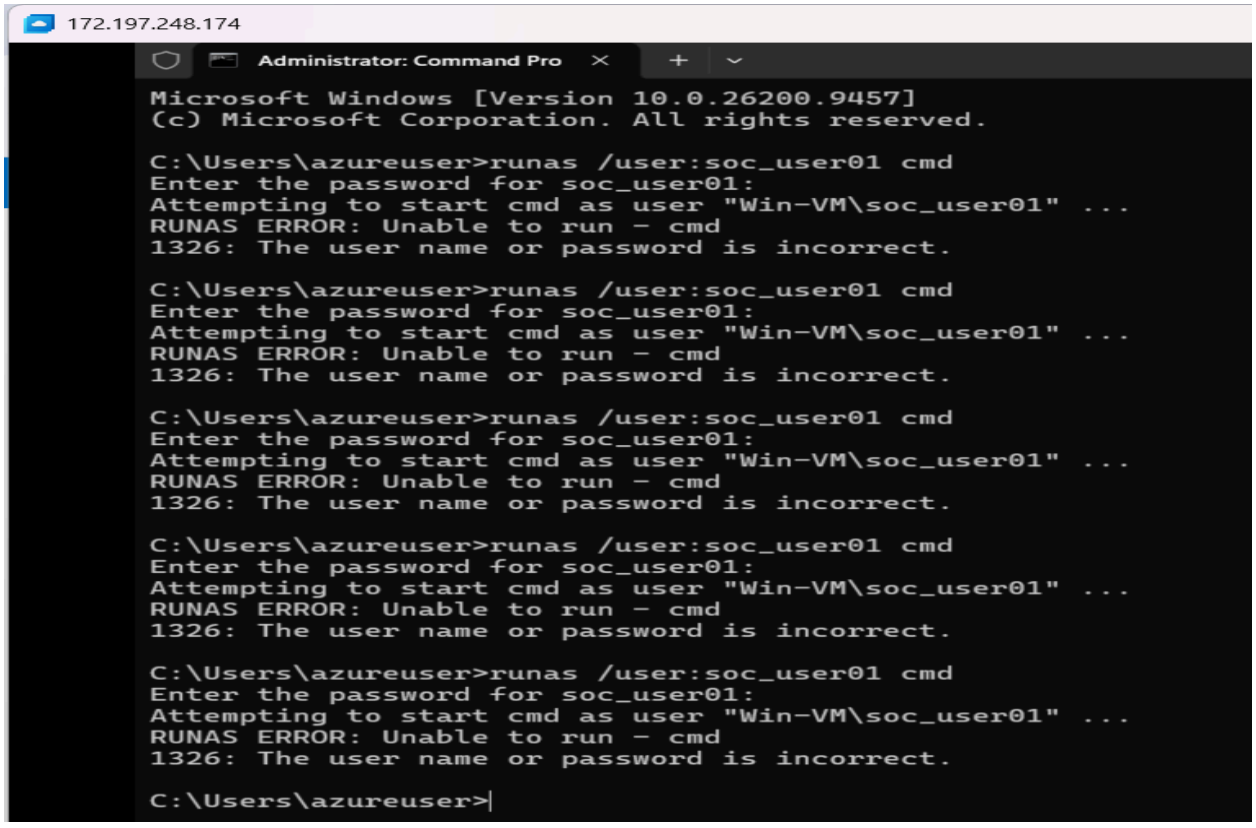
Attack 1 — Failed Login / Brute-Force Simulation

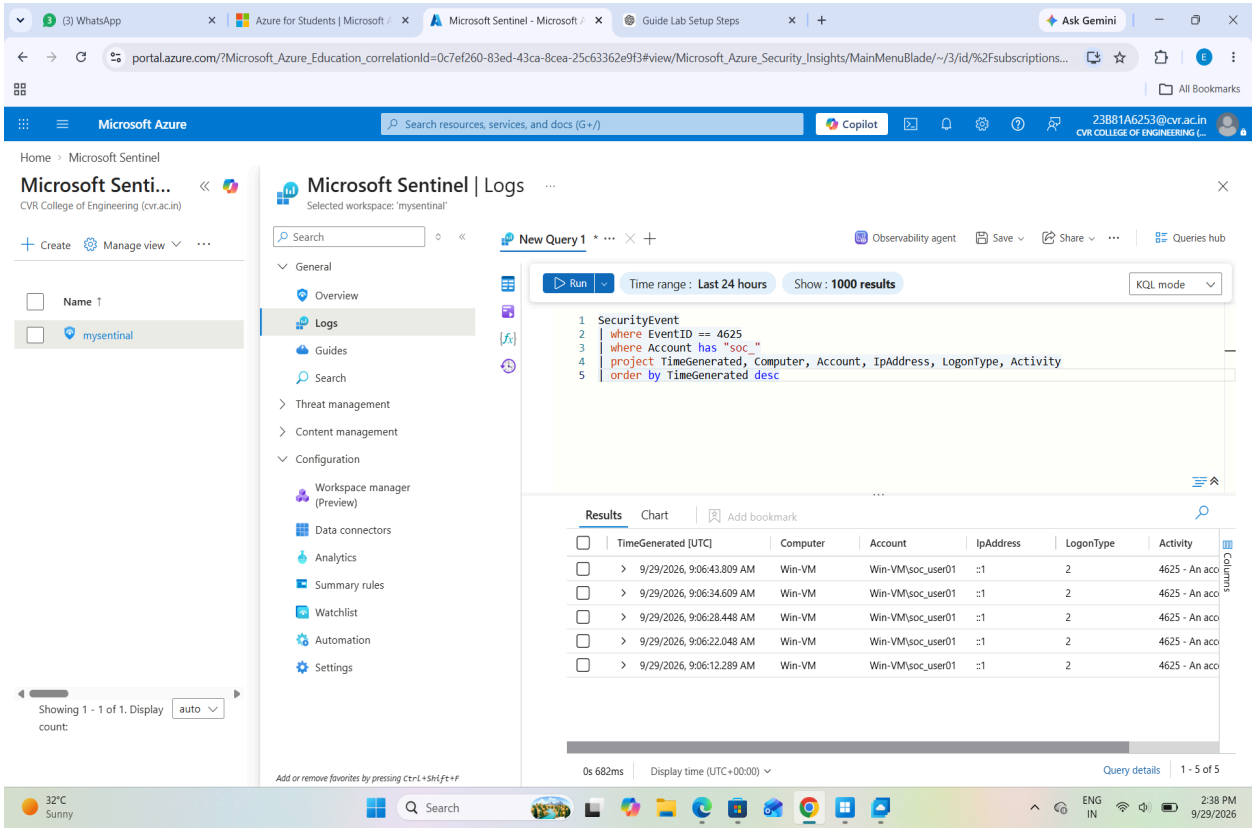
Objective:
In this attack, we simulated multiple failed login attempts against the lab account `soc_user01`. The purpose was to generate failed authentication events that Microsoft Sentinel could detect and investigate. The expected Windows Security Event ID is **4625 – An account failed to log on**.

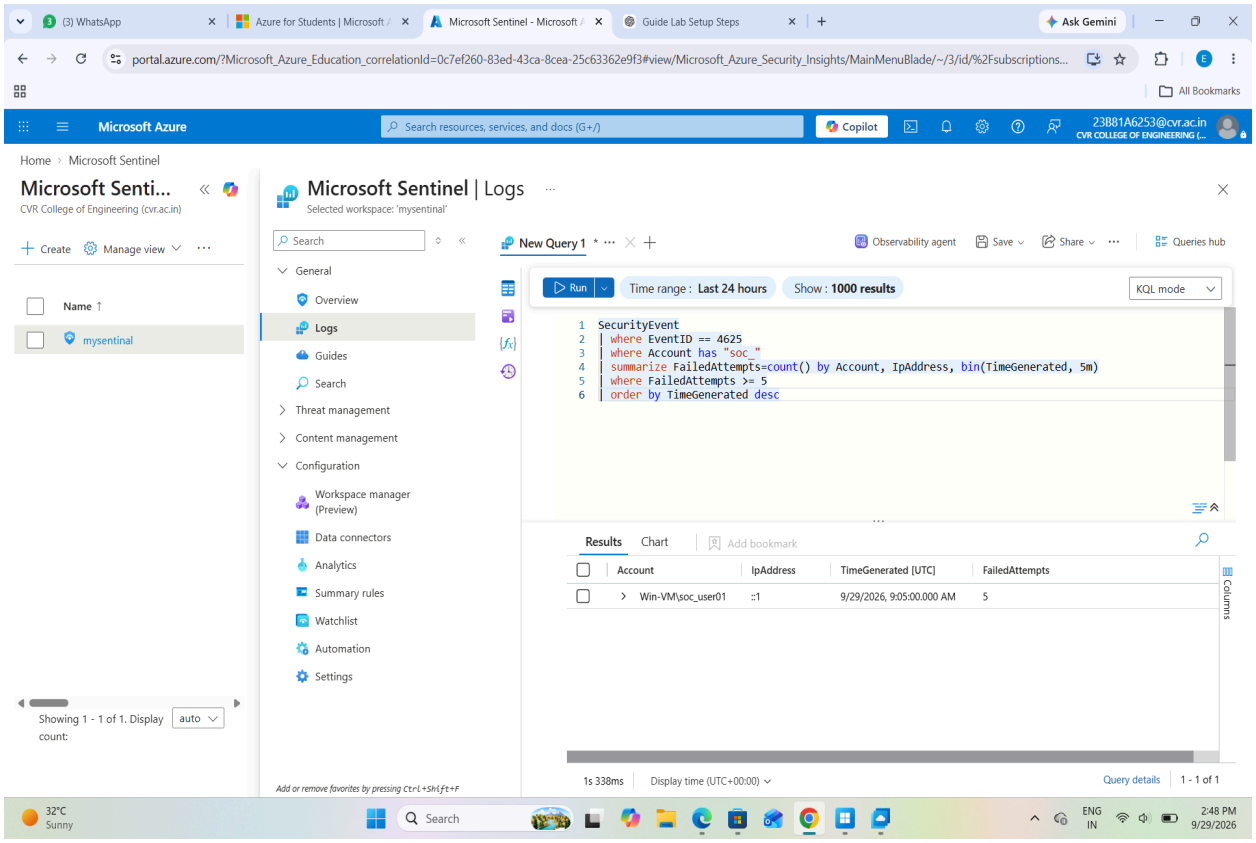
Procedure:
We attempted to log in to the Windows VM using `soc_user01` with an intentionally incorrect password several times. These failed attempts generated security events on the Windows VM.

Sentinel Investigation:
In Microsoft Sentinel → **Logs**, we searched for Event ID **4625** and filtered for the `soc_` accounts. The query displayed the time, computer, account, IP address, logon type, and activity related to the failed attempts.

Detection:
We then grouped the failed login events by account, IP address, and a **5-minute time window**. If 5 or more failures occurred, Sentinel identified it as **potential brute-force activity** requiring further investigation.


172.197.248.174
Administrator: Command Pro
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.
C:\Users\azureuser>


Microsoft Azure
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Copilot
Home > Microsoft Sentinel
Microsoft Senti...
CVR College of Engineering (cvr.ac.in)
Create
Manage view
Name
mysentinal
Showing 1 - 1 of 1. Display count:
auto
Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
Search
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
New Query 1
Observability agent
Save
Share
Queries hub
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
SecurityEvent
| where EventID == 4625
| where Account has "soc_"
| project TimeGenerated, Computer, Account, IpAddress, LogonType, Activity
| order by TimeGenerated desc
Results
Chart
Add bookmark
TimeGenerated [UTC]
Computer
Account
IpAddress
LogonType
Activity
9/29/2026, 9:06:43.809 AM Win-VM Win-VM\soc_user01 ::1 2 4625 - An acc
9/29/2026, 9:06:34.609 AM Win-VM Win-VM\soc_user01 ::1 2 4625 - An acc
9/29/2026, 9:06:28.448 AM Win-VM Win-VM\soc_user01 ::1 2 4625 - An acc
9/29/2026, 9:06:22.048 AM Win-VM Win-VM\soc_user01 ::1 2 4625 - An acc
9/29/2026, 9:06:12.289 AM Win-VM Win-VM\soc_user01 ::1 2 4625 - An acc
Columns
0s 682ms
Display time (UTC+00:00)
Query details
1 - 5 of 5
Add or remove favorites by pressing Ctrl+Shift+F
32°C Sunny
Search
ENG IN
2:38 PM 9/29/2026



Attack 2 – Failed Login → Successful Login

Objective:
To correlate failed and successful login events and understand the sequence of authentication activity.

Procedure:
After generating the failed login attempts in Attack 1, we logged in successfully to the Windows VM using the correct password for **`soc_user01`**.

Sentinel Investigation:
In Microsoft Sentinel, we searched for both **Event ID 4625** and **Event ID 4624** using the following KQL query:

```
SecurityEvent
| where EventID in (4624, 4625)
| where Account has "soc_"
| project TimeGenerated, EventID, Computer, Account, IpAddress, LogonType, Activity
| order by TimeGenerated desc
```

4625 – Failed Login → 4625 – Failed Login → 4625 – Failed Login → 4625 – Failed Login → 4624 – Successful Login

```
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

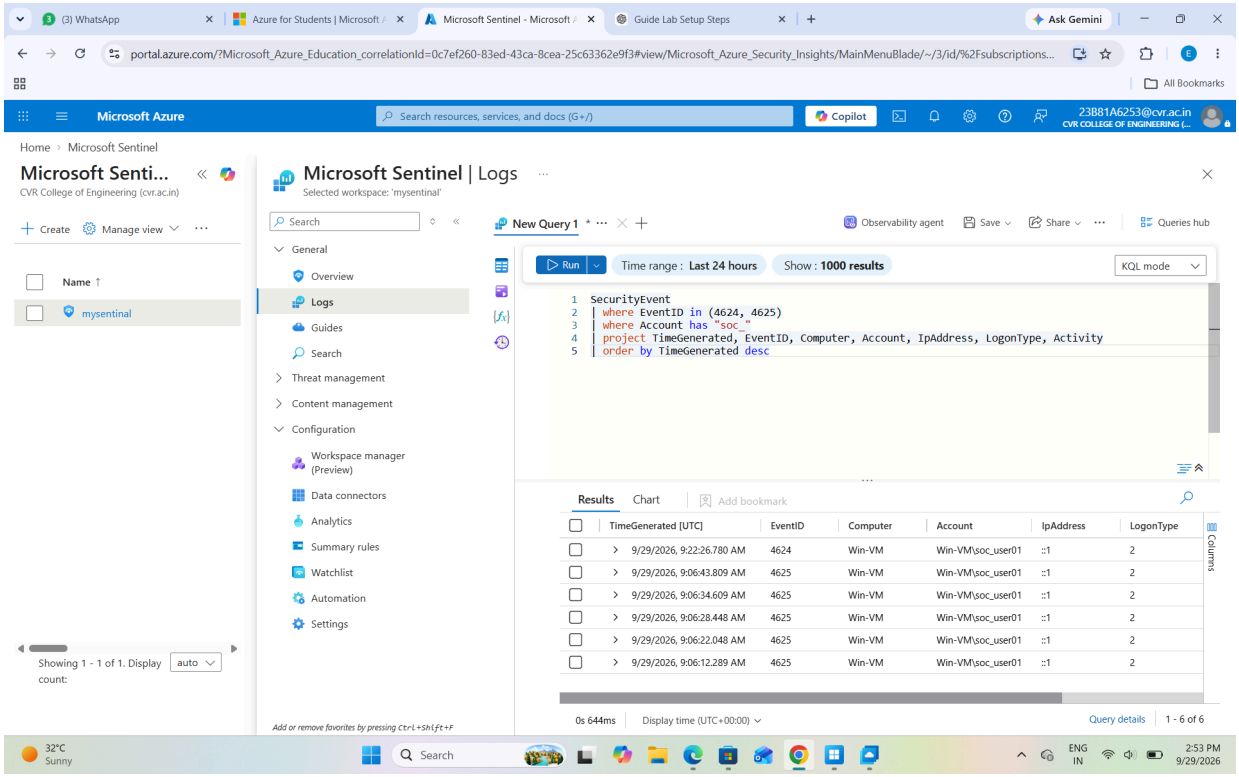
C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
RUNAS ERROR: Unable to run - cmd
1326: The user name or password is incorrect.

C:\Users\azureuser>runas /user:soc_user01 cmd
Enter the password for soc_user01:
Attempting to start cmd as user "Win-VM\soc_user01" ...
```



Attack 3 – Password Spray Simulation

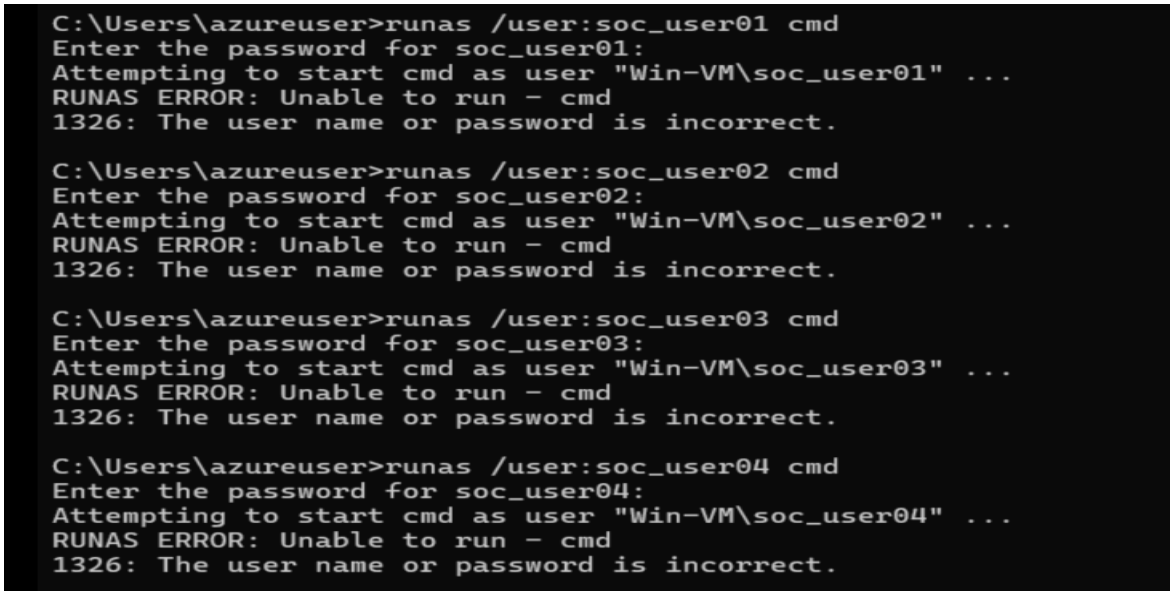
Objective:
To simulate password-spraying activity and detect multiple accounts being targeted from the same source.

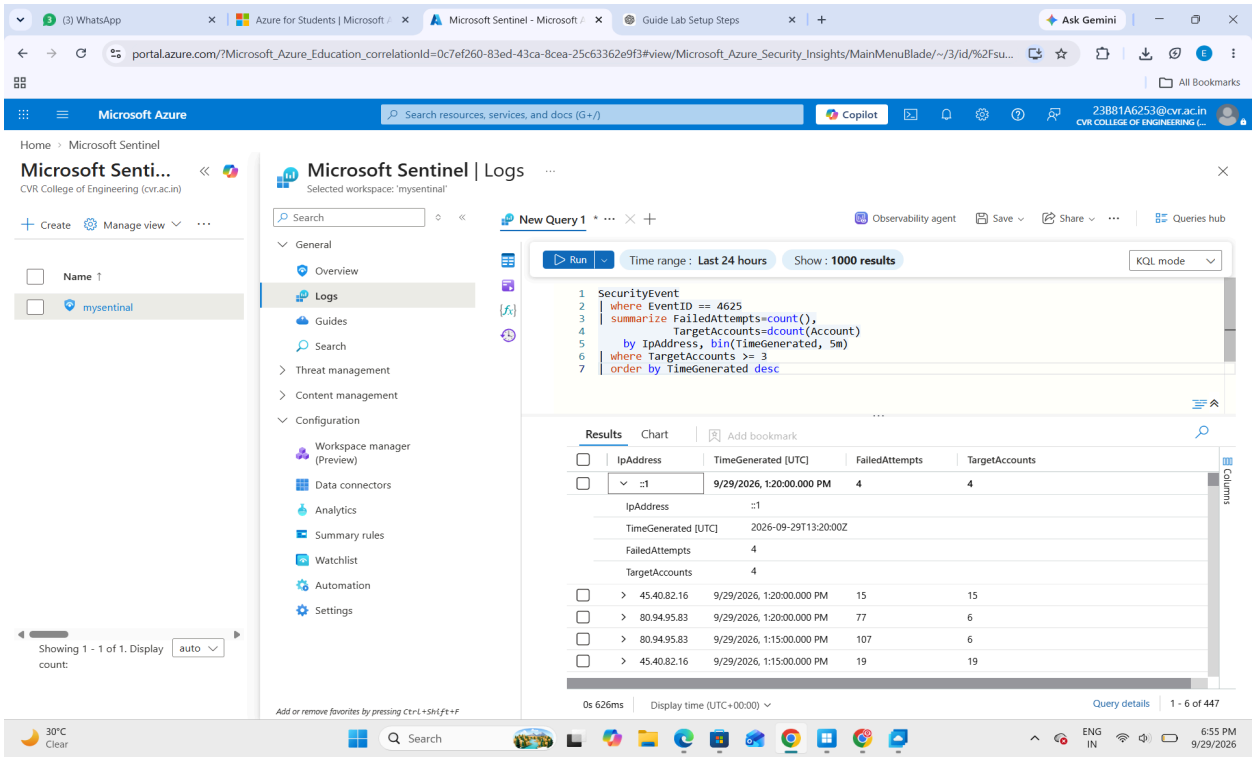
Procedure:
We attempted to log in to **`soc_user01`**, **`soc_user02`**, **`soc_user03`**, and **`soc_user04`** using the same intentionally incorrect password, `WrongPassword@123`.

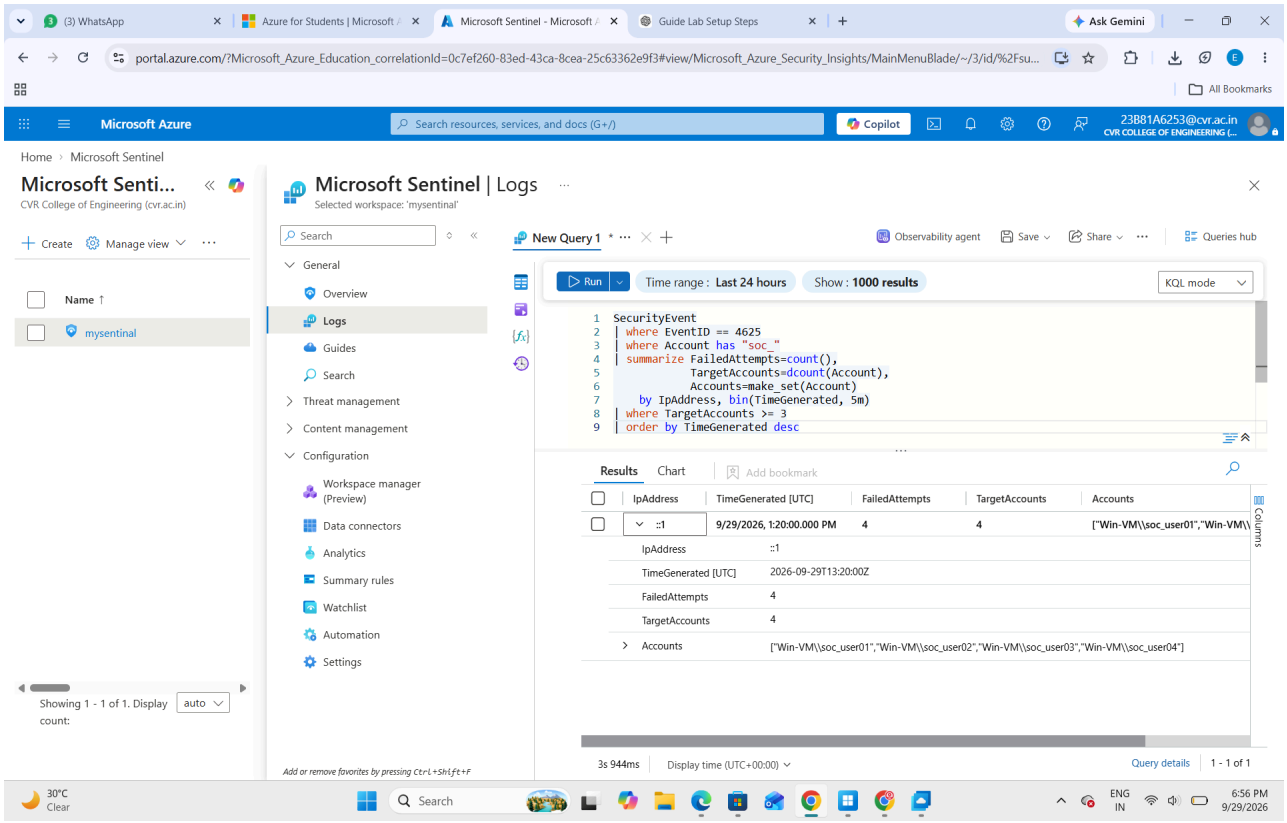
Sentinel Investigation:
In Microsoft Sentinel, we searched for **Event ID 4625** and counted the number of different accounts targeted from the same IP address within a **5-minute window**.

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count(), TargetAccounts=dcount(Account)
    by IpAddress, bin(TimeGenerated, 5m)
| where TargetAccounts >= 3
| order by TimeGenerated desc
```

Observation:
The `dcount(Account)` function shows the number of distinct accounts targeted. If multiple accounts are targeted from the same IP within a short period, it can indicate **potential password-spraying activity**.




Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home › Microsoft Sentinel
Microsoft Senti...
CVR College of Engineering (cvr.ac.in)
Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
New Query 1
Observability agent
Save
Share
Queries hub
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count(),
TargetAccounts=dcount(Account)
by IpAddress, bin(TimeGenerated, 5m)
| where TargetAccounts >= 3
| order by TimeGenerated desc
Results
Chart
Add bookmark
IpAddress
TimeGenerated [UTC]
FailedAttempts
TargetAccounts
::1
9/29/2026, 1:20:00.000 PM
4
4
45.40.82.16
9/29/2026, 1:20:00.000 PM
15
15
80.94.95.83
9/29/2026, 1:20:00.000 PM
77
6
80.94.95.83
9/29/2026, 1:15:00.000 PM
107
6
45.40.82.16
9/29/2026, 1:15:00.000 PM
19
19
0s 626ms
Display time (UTC+00:00)
Query details
1 - 6 of 447
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Name
mysentinal
Showing 1 - 1 of 1. Display count:
auto
Add or remove favorites by pressing Ctrl+Shift+F


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
Home › Microsoft Sentinel
Microsoft Senti...
CVR College of Engineering (cvr.ac.in)
Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
New Query 1
Observability agent
Save
Share
Queries hub
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
SecurityEvent
| where EventID == 4625
| where Account has "soc_"
| summarize FailedAttempts=count(),
TargetAccounts=dcount(Account),
Accounts=make_set(Account)
by IpAddress, bin(TimeGenerated, 5m)
| where TargetAccounts >= 3
| order by TimeGenerated desc
Results
Chart
Add bookmark
IpAddress
TimeGenerated [UTC]
FailedAttempts
TargetAccounts
Accounts
::1
9/29/2026, 1:20:00.000 PM
4
4
["Win-VM\\soc_user01","Win-VM\\soc_user02","Win-VM\\soc_user03","Win-VM\\soc_user04"]
3s 944ms
Display time (UTC+00:00)
Query details
1 - 1 of 1
General
Overview
Logs
Guides
Search
Threat management
Content management
Configuration
Workspace manager (Preview)
Data connectors
Analytics
Summary rules
Watchlist
Automation
Settings
Name
mysentinal
Showing 1 - 1 of 1. Display count:
auto
Add or remove favorites by pressing Ctrl+Shift+F

Attack 4 – Unauthorized Account Creation

Objective:
To simulate the creation of a new user account and detect it using Microsoft Sentinel.

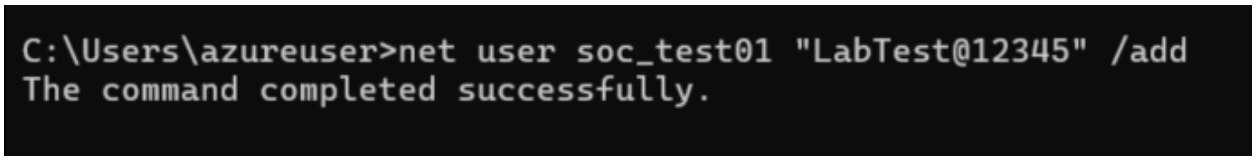
Procedure:
We created a temporary account named `soc_test01` using Administrator Command Prompt:

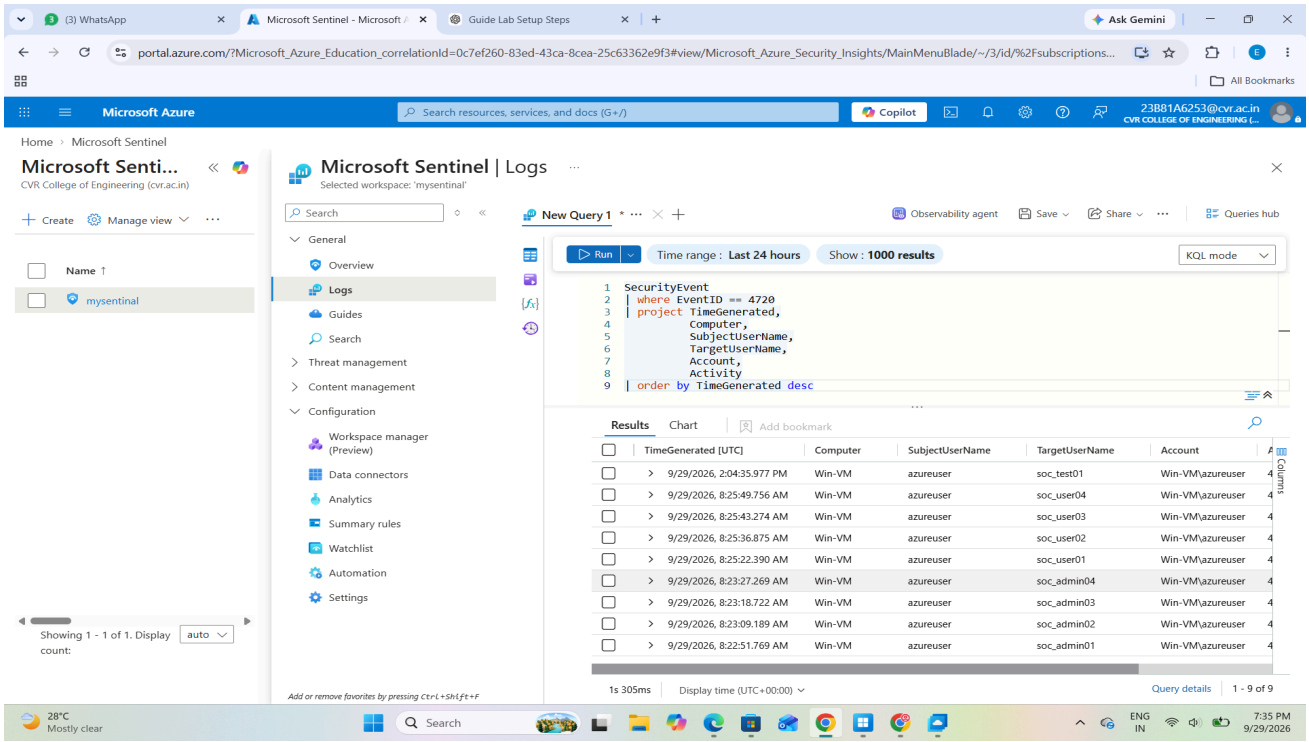
net user soc_test01 "LabTest@12345" /add

This generated Event ID 4720, which represents a user account being created.

Sentinel Investigation:
In Microsoft Sentinel, we searched for Event ID 4720 and checked details such as the time, computer, account that performed the action, and the newly created account.

Observation:
The main investigation question was: “Who created `soc_test01`?”





Attack 5 – Privilege Escalation

Objective:
To simulate a privilege escalation by changing a standard user into a local administrator.

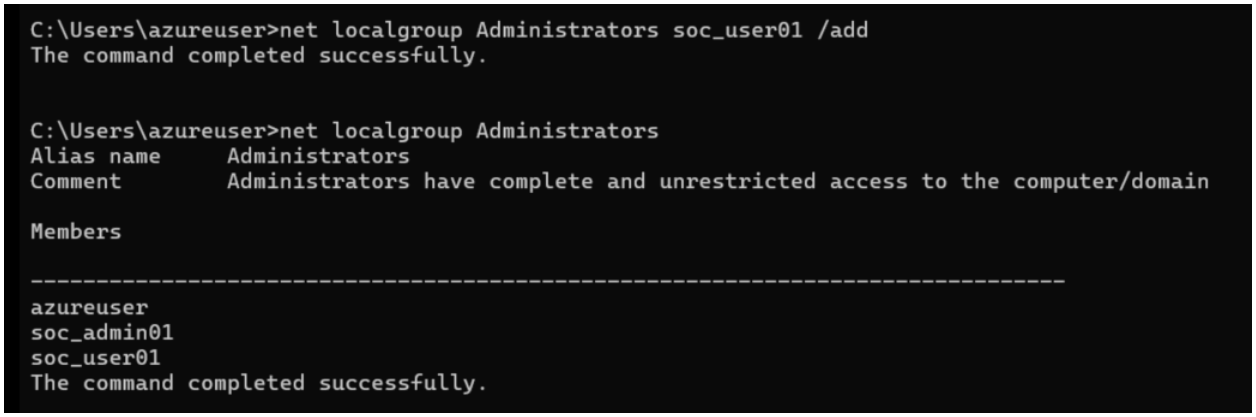
Procedure:
We added `soc_user01`, which was initially a standard user, to the local Administrators group using:

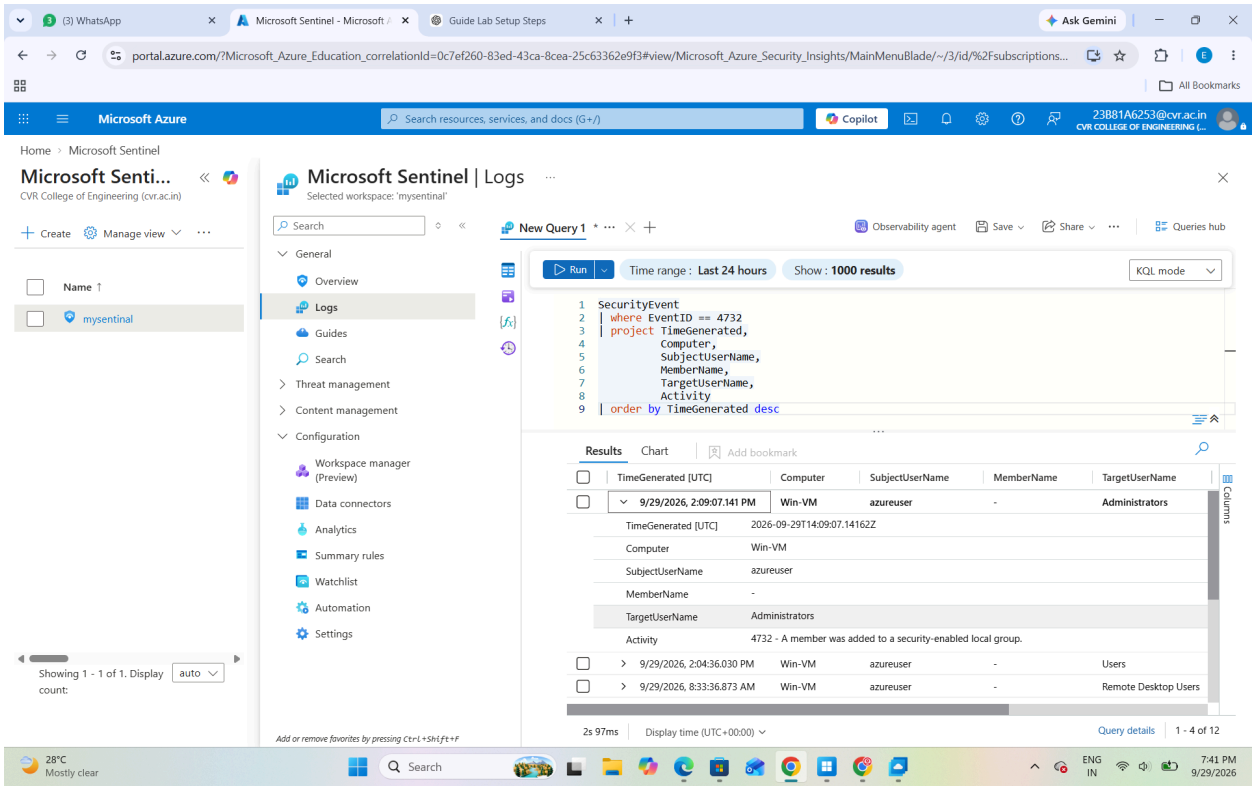
net localgroup Administrators soc_user01 /add

This generated Event ID 4732, which indicates that a member was added to a security-enabled local group.

Sentinel Investigation:
In Microsoft Sentinel, we searched for Event ID 4732 and examined the user who performed the change, the member added, and the target group.

Observation:
The investigation focused on the question: “Why was the standard account `soc_user01` added to the local Administrators group?”





Attack 6 – Privileged Logon

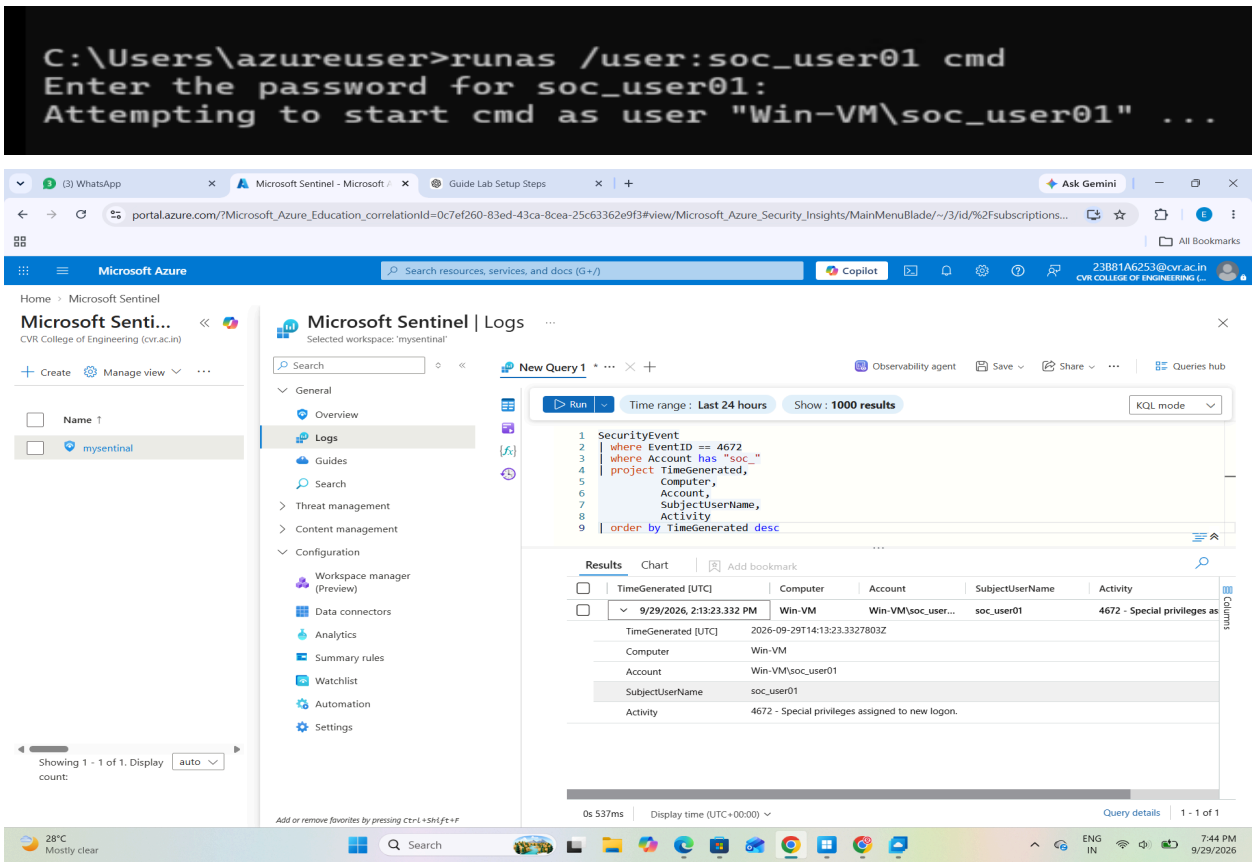
Objective:
To identify a logon where special privileges were assigned to the user.

Procedure:
After adding `soc_user01` to the Administrators group in Attack 5, we logged out and logged in again using `soc_user01`.

Sentinel Investigation:
In Microsoft Sentinel, we searched for Event ID 4672, which indicates that special privileges were assigned to a new logon.

```
SecurityEvent
| where EventID == 4672
| where Account has "soc_"
| project TimeGenerated, Computer, Account, SubjectUserName, Activity
| order by TimeGenerated desc
```

Observation:
The event provides information about account and the privileges assigned during logon.

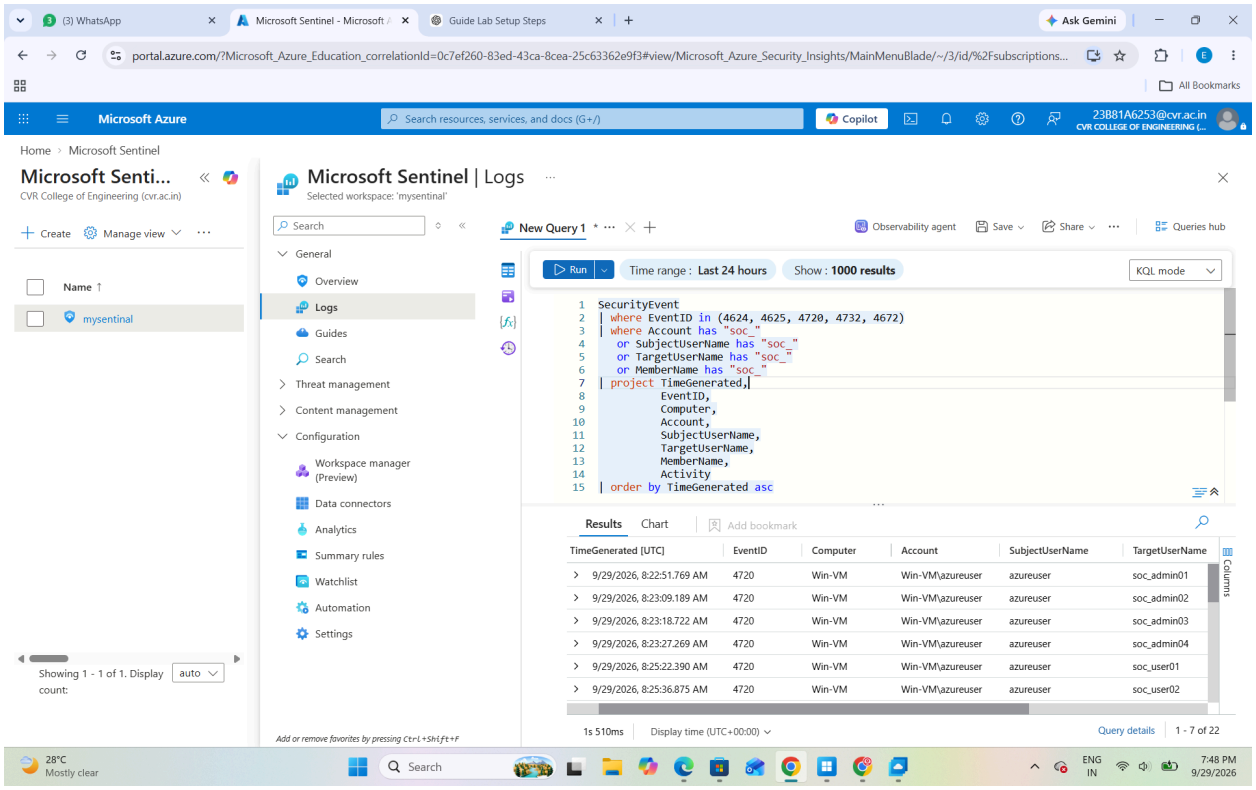


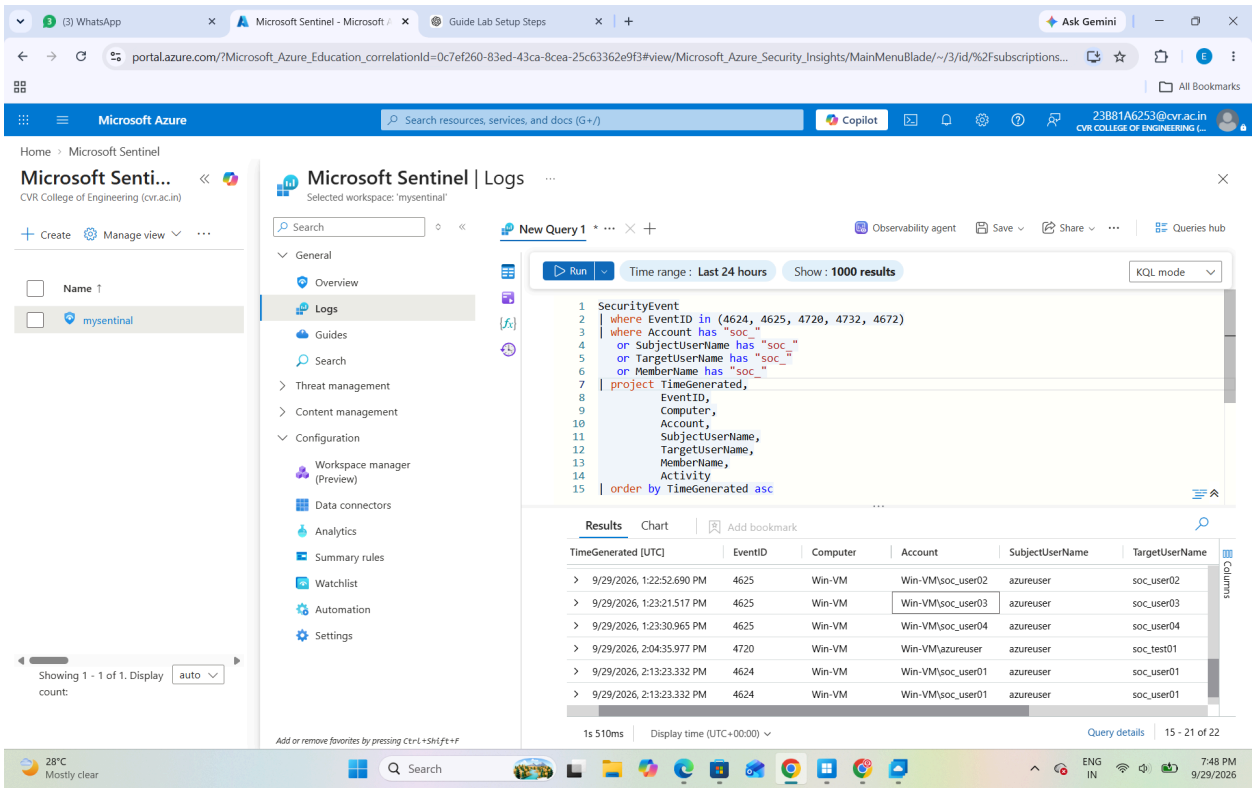
Microsoft Sentinel – Correlated Security Events

This screenshot shows a **Microsoft Sentinel Logs** query used to investigate multiple security events related to the `soc_` lab accounts. The query filters events such as **4624 (successful login), 4625 (failed login), 4720 (account creation), 4732 (group membership change), and 4672 (privileged logon)**.

The results table displays details including the **time, Event ID, computer, account, subject user, and target user**. This helps us correlate different activities performed during the lab and understand the sequence of events.

Observation: The results show **Event ID 4720**, indicating that new user accounts such as `soc_admin01`, `soc_admin02`, `soc_admin03`, `soc_admin04`, `soc_user01`, and `soc_user02` were created on the Windows VM.


Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
New Query 1
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
SecurityEvent
| where EventID in (4624, 4625, 4720, 4732, 4672)
| where Account has "soc_"
or SubjectUserName has "soc_"
or TargetUserName has "soc_"
or MemberName has "soc_"
| project TimeGenerated,
EventID,
Computer,
Account,
SubjectUserName,
TargetUserName,
MemberName,
Activity
| order by TimeGenerated asc
Results
Chart
TimeGenerated [UTC] | EventID | Computer | Account | SubjectUserName | TargetUserName
9/29/2026, 8:22:51.769 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_admin01
9/29/2026, 8:23:09.189 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_admin02
9/29/2026, 8:23:18.722 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_admin03
9/29/2026, 8:23:27.269 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_admin04
9/29/2026, 8:25:22.390 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_user01
9/29/2026, 8:25:36.875 AM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_user02
1s 510ms
Query details
1 - 7 of 22


Microsoft Sentinel | Logs
Selected workspace: 'mysentinal'
New Query 1
Run
Time range : Last 24 hours
Show : 1000 results
KQL mode
SecurityEvent
| where EventID in (4624, 4625, 4720, 4732, 4672)
| where Account has "soc_"
or SubjectUserName has "soc_"
or TargetUserName has "soc_"
or MemberName has "soc_"
| project TimeGenerated,
EventID,
Computer,
Account,
SubjectUserName,
TargetUserName,
MemberName,
Activity
| order by TimeGenerated asc
Results
Chart
TimeGenerated [UTC] | EventID | Computer | Account | SubjectUserName | TargetUserName
9/29/2026, 1:22:52.690 PM | 4625 | Win-VM | Win-VM\soc_user02 | azureuser | soc_user02
9/29/2026, 1:23:21.517 PM | 4625 | Win-VM | Win-VM\soc_user03 | azureuser | soc_user03
9/29/2026, 1:23:30.965 PM | 4625 | Win-VM | Win-VM\soc_user04 | azureuser | soc_user04
9/29/2026, 2:04:35.977 PM | 4720 | Win-VM | Win-VM\azureuser | azureuser | soc_test01
9/29/2026, 2:13:23.332 PM | 4624 | Win-VM | Win-VM\soc_user01 | azureuser | soc_user01
9/29/2026, 2:13:23.332 PM | 4624 | Win-VM | Win-VM\soc_user01 | azureuser | soc_user01
1s 510ms
Query details
15 - 21 of 22

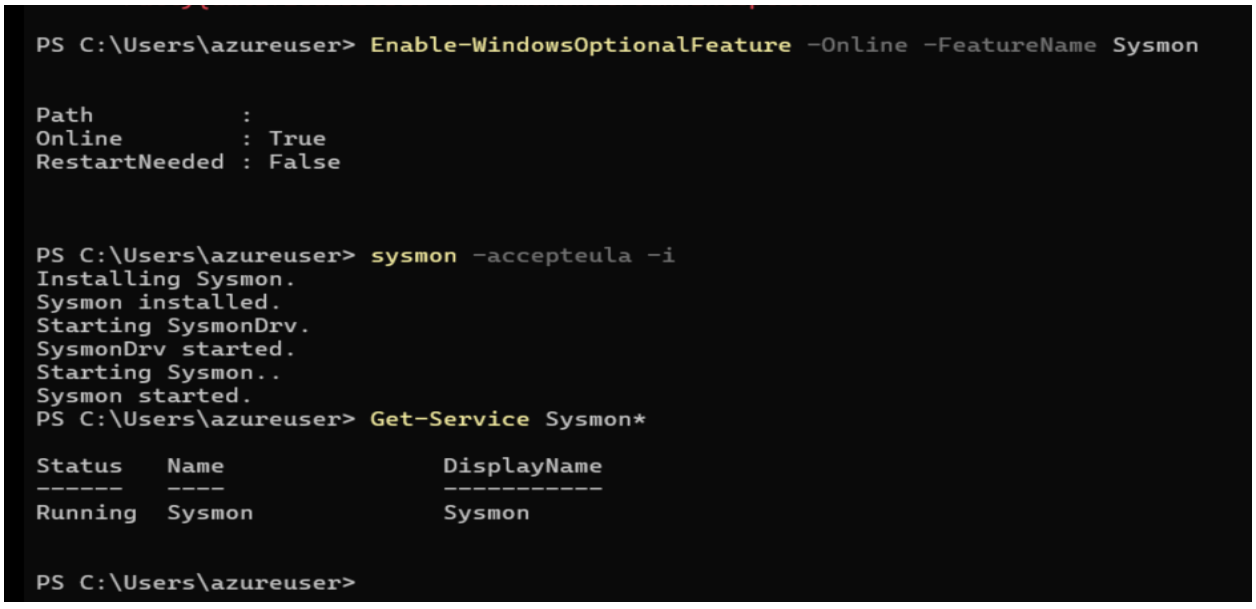
Attack 7 – Process Creation

Objective:
To monitor and investigate newly created processes using Sysmon.

Procedure:
First, we verified that Sysmon was running and checked the Sysmon Operational event log. We then generated some harmless process activity by running Notepad, `whoami`, `ipconfig`, and `Get-Process`.

Investigation:
We searched the Sysmon Operational log for Event ID 1, which records process creation. The event provides useful details such as the process, command line, parent process.

Observation:
The created processes appeared as Process Create events in the Sysmon logs.



```
PS C:\Users\azureuser> Get-Service Sysmon*

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon             Sysmon


PS C:\Users\azureuser> Get-WinEvent -LogName "Microsoft-Windows-Sysmon/Operational" -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
9/29/2026 9:27:58 AM             5 Information      Process terminated:...
9/29/2026 9:26:57 AM             5 Information      Process terminated:...
9/29/2026 9:26:27 AM             1 Information      Process Create:...
9/29/2026 9:26:27 AM             1 Information      Process Create:...
9/29/2026 9:26:24 AM             1 Information      Process Create:...
9/29/2026 9:24:25 AM             1 Information      Process Create:...
9/29/2026 9:24:24 AM             1 Information      Process Create:...
9/29/2026 9:19:53 AM             5 Information      Process terminated:...
9/29/2026 9:16:42 AM             5 Information      Process terminated:...
9/29/2026 9:16:42 AM             5 Information      Process terminated:...

PS C:\Users\azureuser> Get-WinEvent -FilterHashtable @{LogName="Microsoft-Windows-Sysmon/Operational" ; Id=1} -MaxEvents 20


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
9/29/2026 9:26:27 AM             1 Information      Process Create:...
9/29/2026 9:26:27 AM             1 Information      Process Create:...
9/29/2026 9:26:24 AM             1 Information      Process Create:...
9/29/2026 9:24:25 AM             1 Information      Process Create:...
9/29/2026 9:24:24 AM             1 Information      Process Create:...
9/29/2026 9:16:42 AM             1 Information      Process Create:...
9/29/2026 9:16:42 AM             1 Information      Process Create:...
9/29/2026 9:13:44 AM             1 Information      Process Create:...
9/29/2026 9:13:44 AM             1 Information      Process Create:...
9/29/2026 9:09:53 AM             1 Information      Process Create:...
9/29/2026 9:06:57 AM             1 Information      Process Create:...
```

Attack 8 – PowerShell Investigation

Objective:
To investigate PowerShell activity and understand how process telemetry can be used during a SOC investigation.

Procedure:
We ran harmless PowerShell commands such as:

Get-Process
Get-Service

We then investigated the corresponding process-creation events using Sysmon Event ID 1

Investigation:
The investigation focused on identifying:

- Who executed PowerShell
- Which parent process started it
- What command line was executed

- When the process was created

```
PS C:\Users\azureuser> Get-Process

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    140       8     1680       9140       0.31    4480   0 AggregatorHost
    194      14     3528      15980      11.70    2032   0 AMAExtHealthMonitor
    356      19    11920      38836       2.45    1700   2 AppActions
    995      46    22940      69796       2.22   12220   2 backgroundTaskHost
     86       6     3496       5264       0.00    3652   2 cmd
     91       7     3872       5900       0.08    8744   2 cmd
     80       8     1232       6212       0.36    2640   0 conhost
     76       8     1132       6164       0.05    2988   0 conhost
     76       8     1048       6808       0.00    9124   0 conhost
    141       9     1412       9780       0.02    9300   2 conhost
    196      13     2324      18436       0.19   10036   2 conhost
    474      24     8836      28548       0.75    3700   2 CrossDeviceResume
    536      26     2188       7160       3.91     664   0 csrss
    153      12     1932       6220       2.77     744   1 csrss
    443      22     2384       7880       7.47     964   2 csrss
    526      19     5628      29424       2.98    9996   2 ctfmon
     83       7     1152       6340       0.02    1528   2 DefenderSessionHelper
    266      26     5808      19008       0.55    7816   2 dllhost
    169      10     1896      15324       0.31    7932   2 dllhost
    214      15     2964      14528       0.09   10052   2 dllhost
    758      31    24332      34380      27.86    1168   1 dwm
   1235      56    37484     105708      15.44    2868   2 dwm
   3383      90   107096     209632      47.06    4932   2 explorer
```

```
PS C:\Users\azureuser> Get-Service

Status   Name               DisplayName
------   ----               -----------
Stopped  AarSvc_d9016c      Agent Activation Runtime_d9016c
Stopped  ADPSvc             Aggregated Data Platform Service
Stopped  ALG                Application Layer Gateway Service
Running  AppIDSvc           Application Identity
Running  Appinfo            Application Information
Stopped  AppMgmt            Application Management
Stopped  AppReadiness       App Readiness
Stopped  AppVClient         Microsoft App-V Client
Stopped  AppXSvc            AppX Deployment Service (AppXSVC)
Stopped  ApxSvc             Windows Virtual Audio Device Proxy ...
Stopped  AssignedAccessM... AssignedAccessManager Service
Running  AudioEndpointBu... Windows Audio Endpoint Builder
Running  Audiosrv           Windows Audio
Stopped  autotimesvc        Cellular Time
Stopped  AxInstSV           ActiveX Installer (AxInstSV)
Stopped  BcastDVRUserSer... GameDVR and Broadcast User Service_...
Running  BDESVC             BitLocker Drive Encryption Service
Running  BFE                Base Filtering Engine
Stopped  BITS               Background Intelligent Transfer Ser...
Stopped  BluetoothUserSe... Bluetooth User Support Service_d9016c
Running  BrokerInfrastru... Background Tasks Infrastructure Ser...
Stopped  BTAGService        Bluetooth Audio Gateway Service
Stopped  BthAvctpSvc        AVCTP service
Stopped  bthserv            Bluetooth Support Service
```

Attack 9- File Creation

Objective:
To simulate a harmless file-creation activity on the Windows Virtual Machine and verify that Sysmon records the activity.

Steps Performed

1. Open PowerShell

PowerShell was opened on the Windows Virtual Machine.

2. Create a Temporary Directory

The following command was executed:

New-Item C:\Temp -ItemType Directory -Force

```
PS C:\Windows\system32> New-Item C:\Temp -ItemType Directory -Force


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         10/4/2026   5:20 PM                Temp
```

This creates the `C:\Temp` directory if it does not already exist.

3. Create a Test File

A harmless test file was created using:

"Sentinel SOC Lab Test" | Out-File C:\Temp\soc-test.txt

```
PS C:\Windows\system32> "Sentinel SOC Lab Test" | Out-File C:\Temp\soc-test.txt
```

This created the file `soc-test.txt` inside the `C:\Temp` directory.

4. Sysmon Event Generation

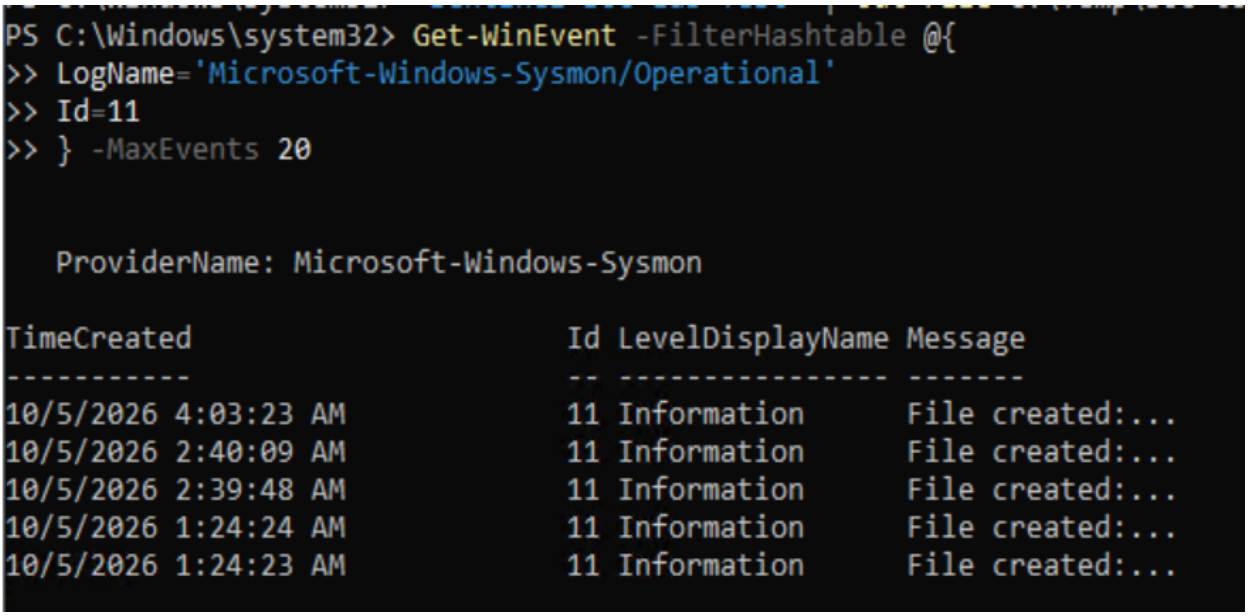
The file-creation activity was recorded by Sysmon Event ID 11, which represents file creation activity.

5. Verify the Event Locally

The following PowerShell command was used:

```
Get-WinEvent -FilterHashtable @{
   LogName='Microsoft-Windows-Sysmon/Operational'
   Id=11
} -MaxEvents 20
```

The output displayed multiple Event ID 11 – Information entries with messages indicating that files were created.



Attack 10 – DNS Investigation

Objective:
To investigate DNS queries and understand how a domain name is resolved to an IP address.

Procedure:
We ran the following command in PowerShell:

nslookup example.com

The command returned the DNS server information, the domain name, and the IP addresses resolved for `example.com`.

Investigation:
Sysmon Event ID 22 records DNS query activity. The investigation follows this chain:

Process → DNS Query → Domain → Resolved IP → Network Connection

```
PS C:\Users\azureuser> nslookup example.com
Server:  UnKnown
Address:  168.63.129.16

Non-authoritative answer:
Name:    example.com
Addresses:  2606:4700:10::ac42:93f3
          2606:4700:10::6814:179a
          172.66.147.243
          104.20.23.154

PS C:\Users\azureuser> nslookup google.com
Server:  UnKnown
Address:  168.63.129.16

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4001:c01::64
          2404:6800:4001:c01::8b
          2404:6800:4001:c01::71
          2404:6800:4001:c01::8a
          173.194.41.139
          173.194.41.138
          173.194.41.113
          173.194.41.102
          173.194.41.101
          173.194.41.100
```

Attack 11 - Test Network Connectivity

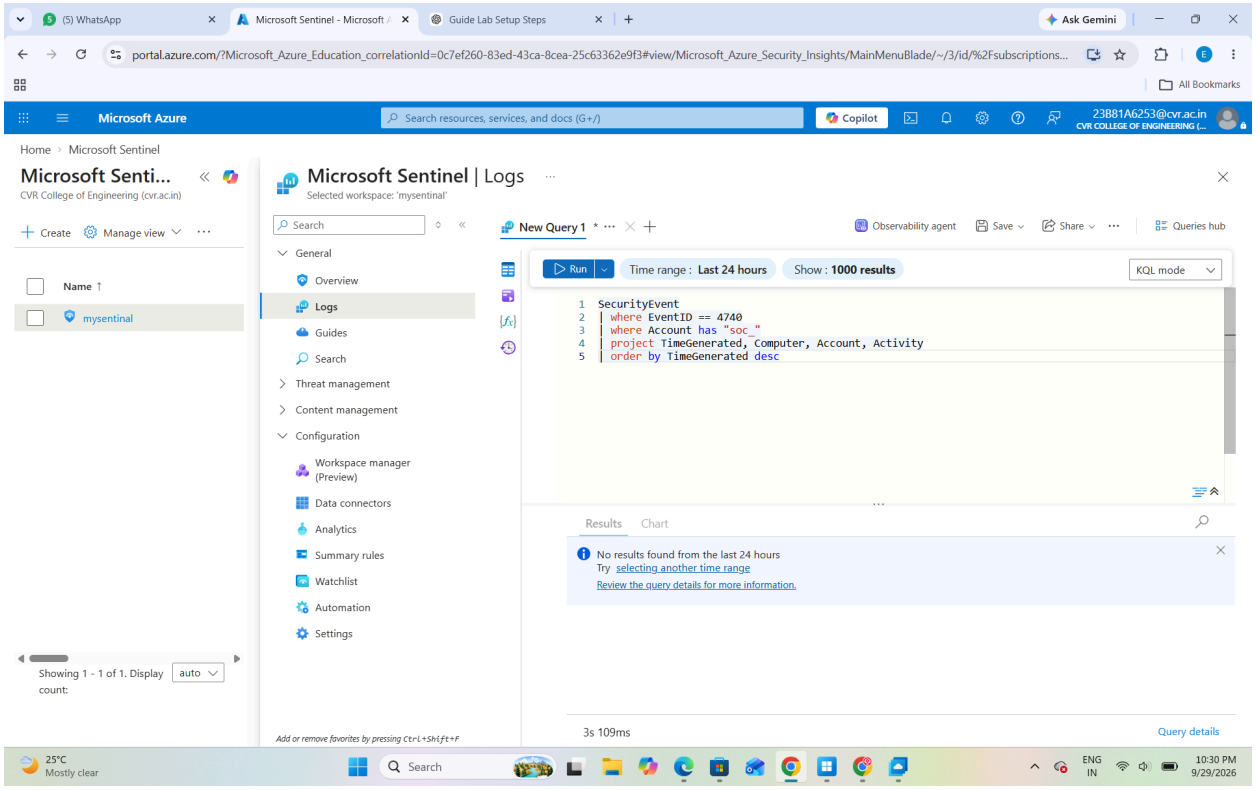
In this task, we used the PowerShell command `Test-NetConnection` to check whether the Windows VM could establish a network connection with `example.com` on port 443 (HTTPS).

The output shows:

- Remote Address: `104.20.23.154`
- Remote Port: `443`
- Interface: Ethernet
- Source Address: `172.16.0.4`
- TcpTestSucceeded: `True`

```
PS C:\Users\azureuser> Test-NetConnection example.com -Port 443


ComputerName     : example.com
RemoteAddress    : 104.20.23.154
RemotePort       : 443
InterfaceAlias   : Ethernet
SourceAddress    : 172.16.0.4
TcpTestSucceeded : True
```



Attack 12 — Registry Modification

Objective

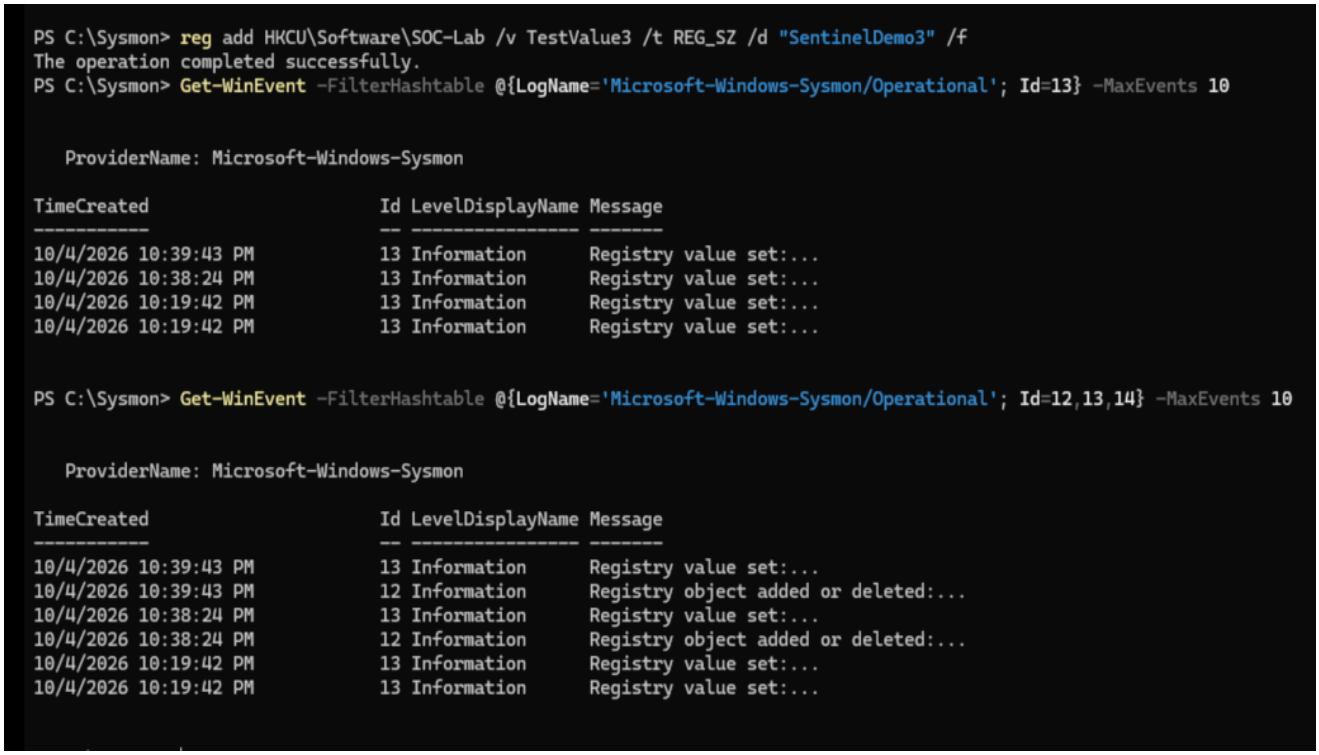
Create a harmless registry entry in the Windows VM and observe the registry modification through Sysmon.

1. Create the test registry key

Open PowerShell/CMD as Administrator and run:

reg add HKCU\Software\SOC-Lab /v TestValue /t REG_SZ /d "SentinelDemo" /f

In Attack 12, I simulated a registry modification by creating a harmless test registry key called SOC-Lab. Sysmon can monitor registry activity through Event IDs 12 to 14. In a real SOC environment, registry modifications are important because attackers may modify registry settings for persistence or configuration changes. Here, we performed only a harmless test modification and investigated the resulting telemetry."



```
PS C:\Sysmon> reg add HKCU\Software\SOC-Lab /v TestValue3 /t REG_SZ /d "SentinelDemo3" /f
The operation completed successfully.
PS C:\Sysmon> Get-WinEvent -FilterHashtable @{LogName='Microsoft-Windows-Sysmon/Operational'; Id=13} -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/4/2026 10:39:43 PM           13 Information      Registry value set:...
10/4/2026 10:38:24 PM           13 Information      Registry value set:...
10/4/2026 10:19:42 PM           13 Information      Registry value set:...
10/4/2026 10:19:42 PM           13 Information      Registry value set:...


PS C:\Sysmon> Get-WinEvent -FilterHashtable @{LogName='Microsoft-Windows-Sysmon/Operational'; Id=12,13,14} -MaxEvents 10


   ProviderName: Microsoft-Windows-Sysmon

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/4/2026 10:39:43 PM           13 Information      Registry value set:...
10/4/2026 10:39:43 PM           12 Information      Registry object added or deleted:...
10/4/2026 10:38:24 PM           13 Information      Registry value set:...
10/4/2026 10:38:24 PM           12 Information      Registry object added or deleted:...
10/4/2026 10:19:42 PM           13 Information      Registry value set:...
10/4/2026 10:19:42 PM           13 Information      Registry value set:...
```

Attack 13 — Account Lockout

Objective

Generate incorrect login attempts on a lab account until the Windows account-lockout policy is triggered.

1. Use a lab account

Use only your test account, for example: `testuser`

Do not use your Azure/administrator account.

2. Generate incorrect login attempts

Enter an incorrect password several times for the test account. The exact number depends on your Windows Account Lockout Policy.

The goal is to generate an account lockout event.

3. Check in Microsoft Sentinel

Go to: Microsoft Sentinel → Logs

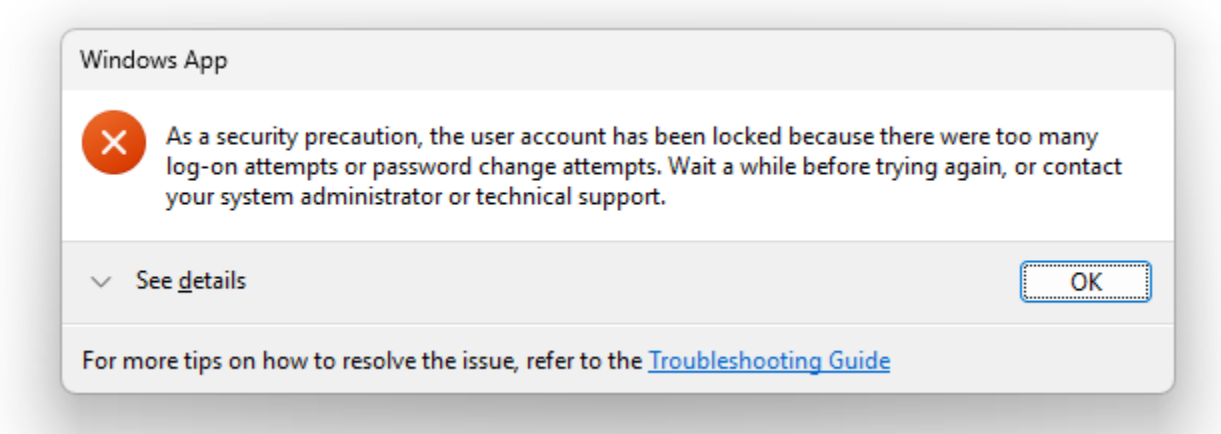
```
SecurityEvent
| where EventID == 4740
| where Account has "soc_"
| project TimeGenerated, Computer, Account, Activity
| order by TimeGenerated desc
```

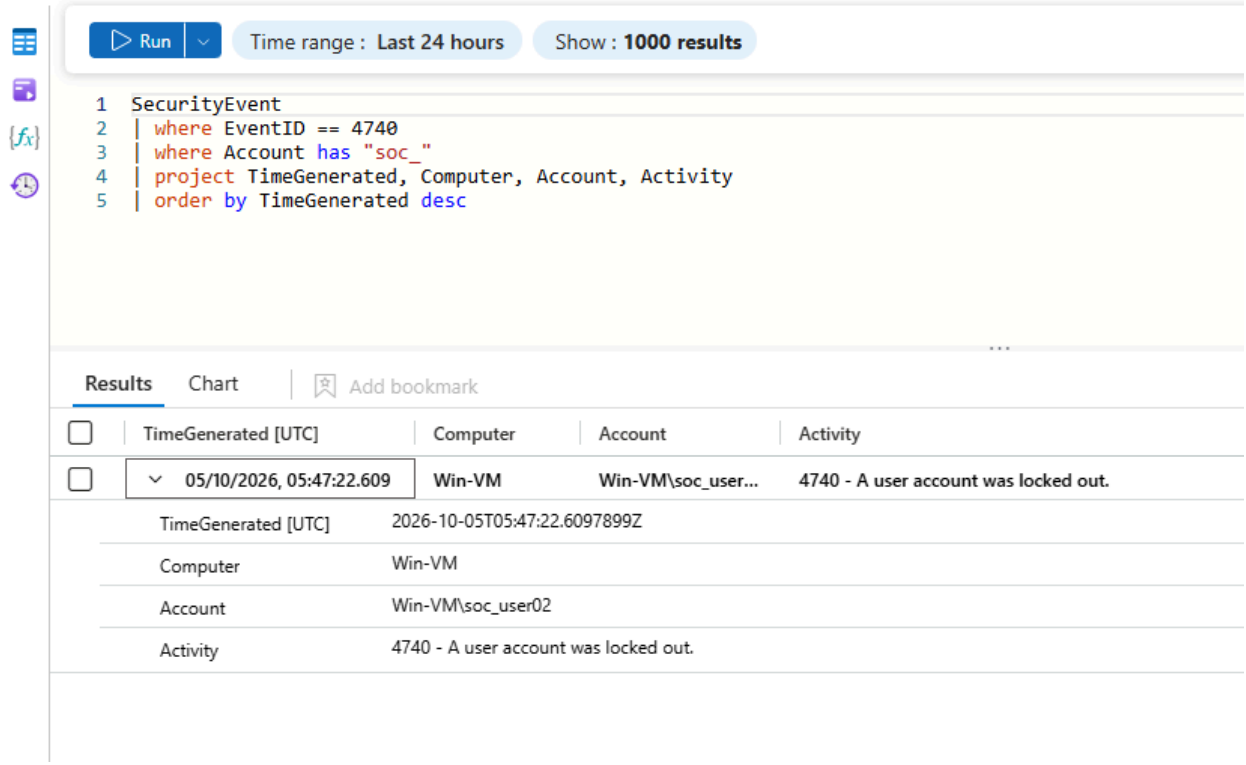
Important Event ID

Event ID 4740 = Account was locked out. MS Sentinel(2)

So the flow is:

Wrong password attempts
↓
Account reaches lockout threshold
↓
Windows generates Event ID 4740
↓
Sentinel collects the event
↓
KQL identifies the locked account





Attack 14 — Scheduled Task

Objective

Create a harmless scheduled task on the Windows VM and detect its creation through Microsoft Sentinel.

1. Create the scheduled task

Open PowerShell as Administrator and run:

schtasks /create /sc once /tn "SOC-Lab-Test" /tr "notepad.exe" /st 23:59

```
PS C:\Sysmon> schtasks /create /sc once /tn "SOC-Lab-Test" /tr "notepad.exe" /st 23:59
SUCCESS: The scheduled task "SOC-Lab-Test" has successfully been created.
PS C:\Sysmon> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4698} -MaxEvents 10


   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
10/4/2026 11:06:51 PM         4698 Information      A scheduled task was created....
```

This creates a task named SOC-Lab-Test that is configured to run `notepad.exe` once at the specified time.

2. Expected Windows Event

The task creation generates:

Event ID 4698 — A scheduled task was created. MS Sentinel(3)

3. Check it in Microsoft Sentinel

Go to: Microsoft Sentinel → Logs

```
SecurityEvent
| where EventID == 4698
| project TimeGenerated, Computer, SubjectUserName, Activity
| order by TimeGenerated desc
```

You should see the scheduled-task creation event.

4. SOC Investigation

The important question is:

> Who created the scheduled task and when was it created?

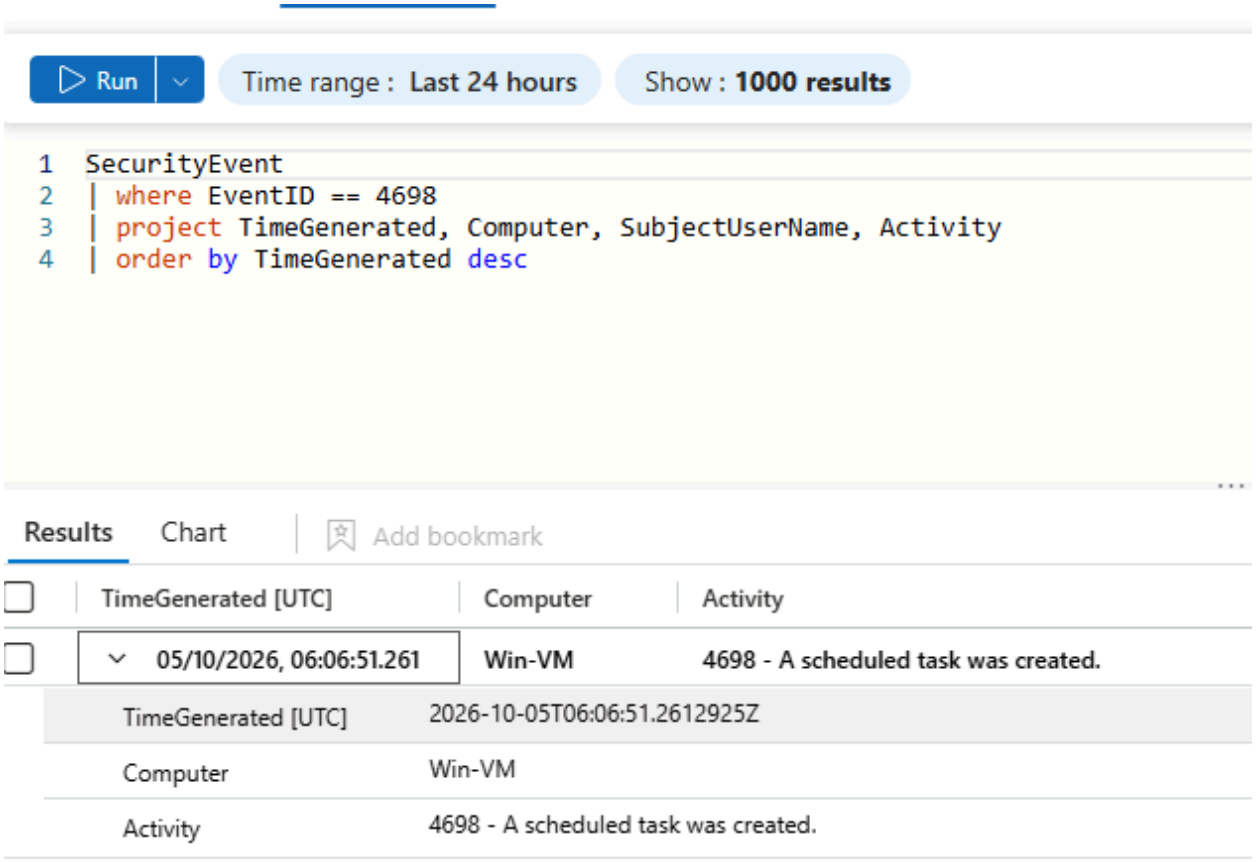
Look at:

- `TimeGenerated` → When it happened
- `SubjectUserName` → Account that created it
- `Computer` → Which machine
- `Activity` → Event description

5. Delete the test task

After the demonstration, remove the task:

```
schtasks /delete /tn "SOC-Lab-Test" /f
```

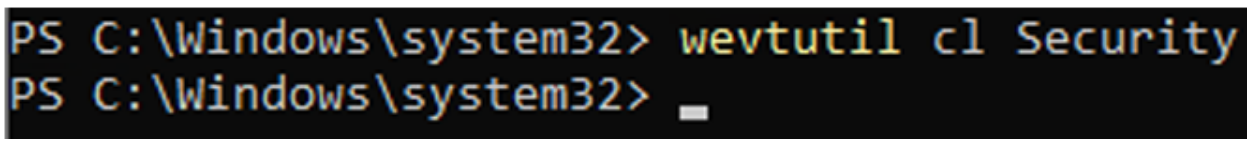


Attack 15 - Security Log Clearing

Objective:
Demonstrate how an attempt to clear the Windows Security log generates its own security event.

Steps:

Open PowerShell as Administrator.

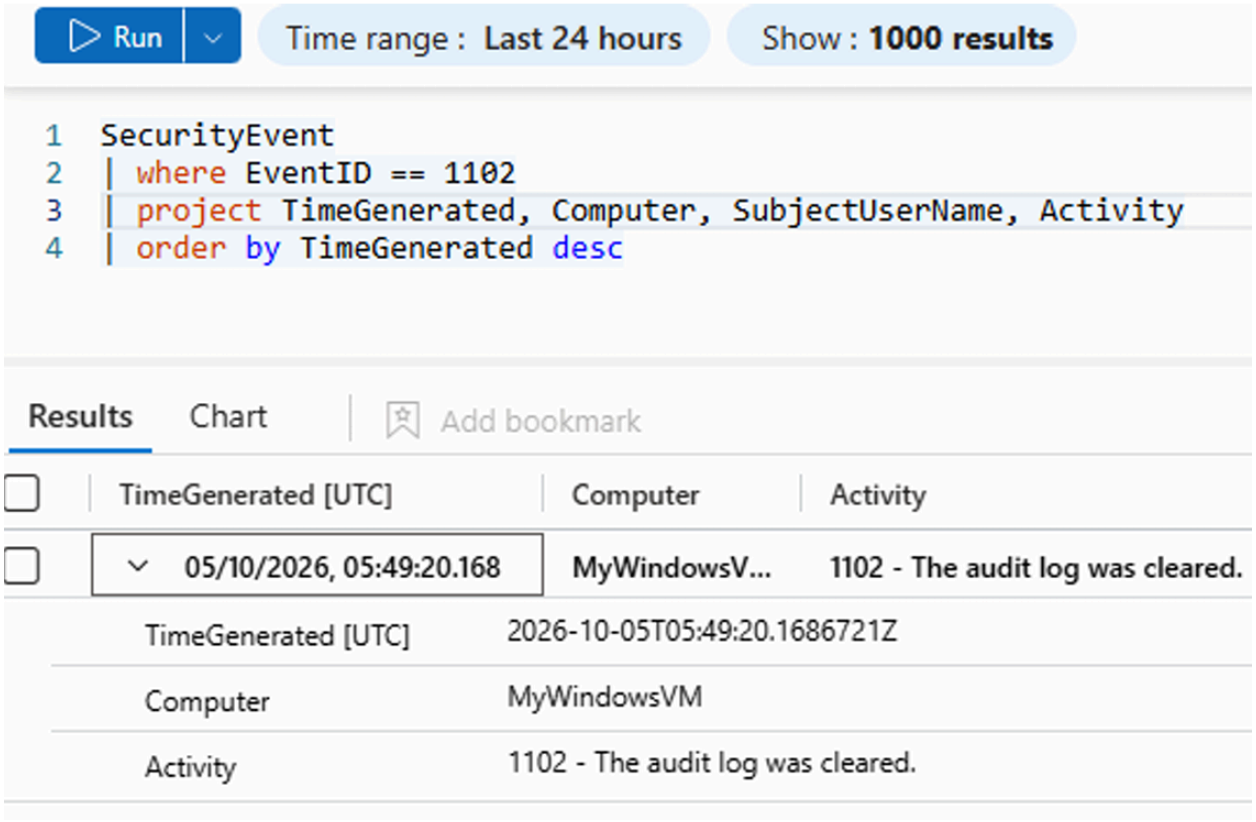


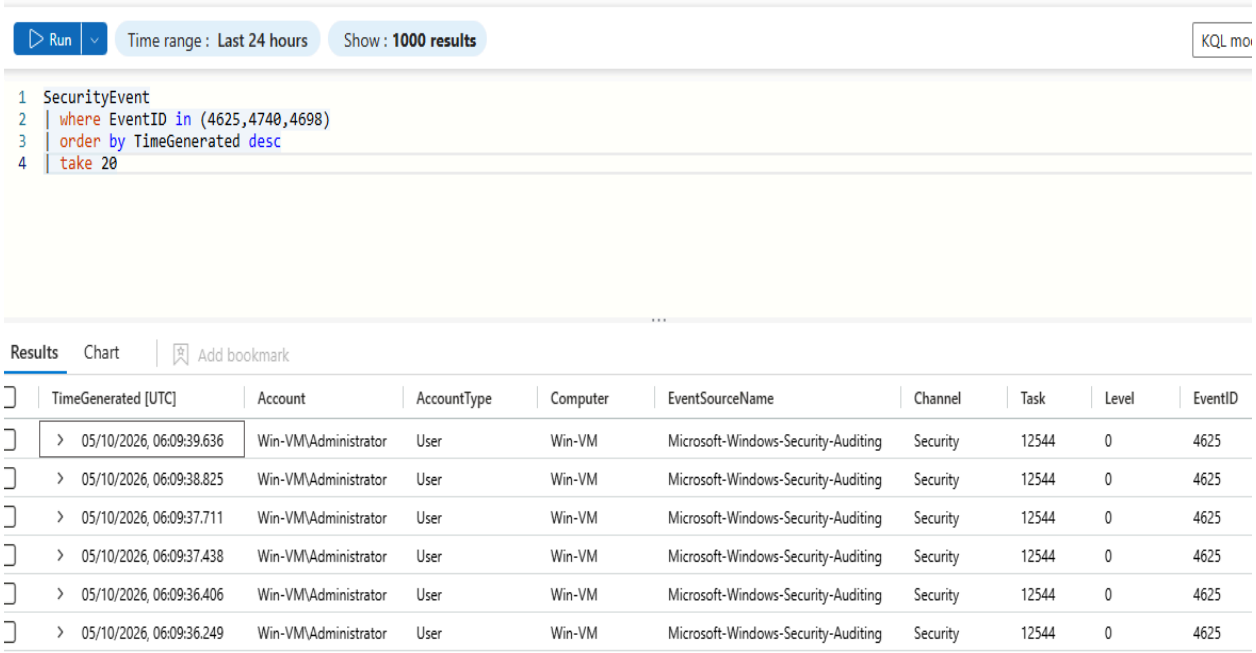
This clears the Windows Security Event Log on the isolated lab VM.
Windows records the log-clearing activity as Event ID 1102.
Verify the event locally using:

```
Get-WinEvent -FilterHashtable @{
   LogName='Security'
   Id=1102
} -MaxEvents 5 | Format-List TimeCreated, Id, Message
```

```
PS C:\Windows\system32> Get-WinEvent -FilterHashtable @{
>> LogName='Security'
>> Id=1102
>> } -MaxEvents 10 | Format-List TimeCreated, Id, Message


TimeCreated : 10/5/2026 5:49:20 AM
Id          : 1102
Message     : The audit log was cleared.
              Subject:
                Security ID:    S-1-5-21-2322504839-385044276-2162829752-500
                Account Name:   azureuser
                Domain Name:    MyWindowsVM
                Logon ID:       0x4659B19
```





Creating Microsoft Sentinel Analytics Rule

1. General Settings

- Rule Name: `SOC Lab - Multiple Failed Logins`
- Description: Detects multiple failed Windows login attempts within a short time period.
- Severity: Medium
- Status: Enabled

2. Set Rule Logic

The following KQL query is used:

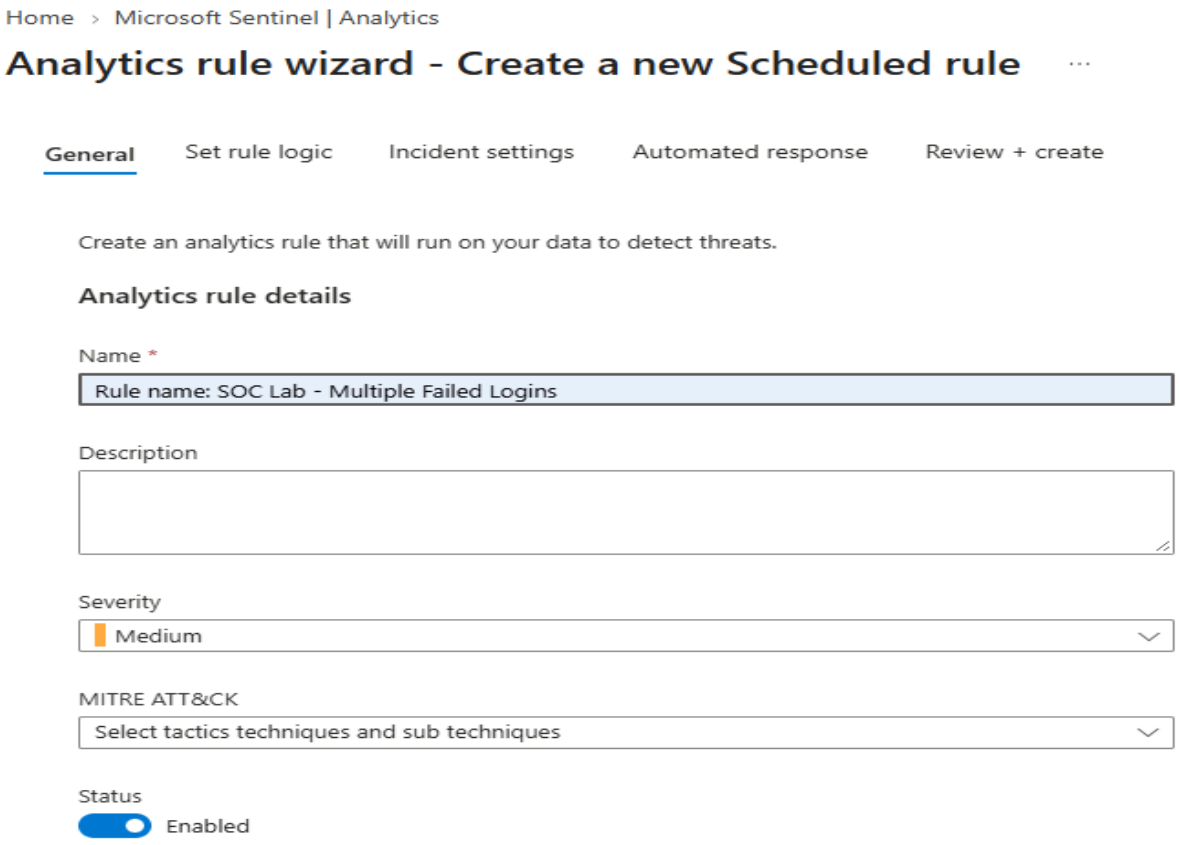
```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count()
    by Account, IpAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 5
| order by TimeGenerated desc
```

Explanation:
The query searches for Windows Event ID 4625, which represents a failed login. It counts failed attempts for the same account and IP address within a 5-minute window. If the number of failures reaches 5 or more, the activity is detected as a potential brute-force login attempt.

3. Query Scheduling

- Run query every: 5 minutes
- Lookup data from the last: 5 minutes
- Start running: Automatically

This allows Microsoft Sentinel to periodically execute the detection query against the collected `SecurityEvent` **data.**



General | **Set rule logic** | Incident settings | Automated response | Review + create

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
SecurityEvent
    by Account, IpAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 5
| order by TimeGenerated desc
```

View query results >

Alert enhancement

- Entity mapping
- Custom details
- Alert details

Query scheduling

Run query every *

5 | Minutes

Lookup data from the last *

5 | Minutes

Start running ⓘ

- (•) Automatically
- ( ) At specific time (Preview)

06/10/2026 | 12:00

< Previous | Next : Incident settings >

1a. Identify and explain the purpose and availability of the Microsoft Sentinel, Log Analytics Workspace, Azure Monitor Agent (AMA), Data Collection Rule, and KQL tools used in the SOC attack simulation. Explain how each tool supports security monitoring and ethical security investigation.

The Microsoft Sentinel–Based SOC Attack Simulation project uses five important tools for security monitoring and investigation.

1. Microsoft Sentinel: Microsoft Sentinel is a cloud-based Security Information and Event Management (SIEM) and Security Orchestration, Automation, and Response (SOAR) solution. It detects suspicious activities, generates alerts, and helps security analysts investigate incidents.

2. Log Analytics Workspace: It is a centralized storage location for collecting and analyzing logs from monitored resources. In this project, it stores Windows security events generated during the simulated attacks.

3. Azure Monitor Agent (AMA): AMA collects configured security and monitoring data from the Windows virtual machine and forwards it to the designated Log Analytics Workspace.

4. Data Collection Rule (DCR): A DCR specifies which event data should be collected, how it should be collected, and where it should be sent. It ensures that relevant Windows logs are available for investigation.

5. Kusto Query Language (KQL): KQL is used to search, filter, summarize, and correlate log data. In this project, it helps identify failed login attempts (Event ID 4625), account creation (Event ID 4720), and administrator-group modifications (Event ID 4732).

How these tools work together: The Windows VM generates security events, AMA collects the configured events according to the DCR, and the Log Analytics Workspace stores them. Microsoft Sentinel analyzes this data using KQL-based analytics rules to detect suspicious behavior and generate alerts for investigation.

These tools support ethical security investigation by allowing students to simulate attacks in an isolated laboratory environment, analyze evidence, identify potential threats, and recommend appropriate response actions.

1b . Explain the business and security implications of the simulated activities, including authentication failures, password-spray activity, account creation, privilege modification, process and file activity, and DNS/network activity. How can Microsoft Sentinel help organizations address these security risks?

The simulated security activities demonstrate how different attacks can affect an organization's security and how Microsoft Sentinel helps detect them.

1. Authentication Failures: Repeated failed login attempts may indicate incorrect passwords or brute-force attacks. Monitoring these events helps identify possible unauthorized access.

2. Password-Spray Activity: Attackers may try a common password against multiple accounts. Detecting failures across several accounts helps identify potential password-spray attacks.

3. Account Creation: Unauthorized account creation may allow attackers to gain or maintain access to a system. Monitoring account-creation events helps identify suspicious changes.

4. Privilege Modification: Adding a standard user to the Administrators group grants elevated permissions. If unauthorized, this can expose sensitive data and allow critical system changes.

5. Process and File Activity: Unexpected processes or file creation may indicate suspicious execution or unauthorized modifications. Investigating these events helps analysts understand system activity.

6. DNS and Network Activity: Unusual DNS queries and network connections may indicate communication with suspicious destinations. Monitoring this activity helps identify potential threats.

Role of Microsoft Sentinel: Microsoft Sentinel analyzes security events, correlates related activities, and generates alerts through Analytics Rules. It helps security analysts investigate incidents and recommend appropriate responses. However, an individual event does not automatically prove an attack; its context must be examined before taking action.

1c. Using Kusto Query Language (KQL) in Microsoft Sentinel, correlate the security events generated during the attack simulations and construct an incident timeline. Identify the affected account, host, sequence of activities, and potential security risks.

Microsoft Sentinel uses KQL to investigate and correlate security events generated on the Windows virtual machine. By analyzing event IDs and timestamps, a SOC analyst can reconstruct the sequence of activities and identify suspicious behavior.

1. KQL query to construct the incident timeline

```
SecurityEvent
| where Account has "soc_"
| where EventID in (4624, 4625, 4648, 4672, 4688,
                    4697, 4720, 4728, 4732, 4740, 1102)
| project TimeGenerated, EventID, Computer, Account,
    SubjectUserName, TargetUserName, Activity
| order by TimeGenerated asc
```

Event ID	Activity	Security Significance
4625	Failed login	May indicate brute-force attempts.
4624	Successful login	Identifies a successful authentication.
4672	Special privileges assigned	Indicates a privileged logon.
4688	Process creation	Helps investigate process execution.
4720	User account created	May indicate unauthorized account creation.
4732	Member added to a local security group	May indicate privilege escalation.
4740	Account locked out	May indicate repeated failed logins.
1102	Security audit log cleared	May indicate an attempt to remove evidence.

3. Incident timeline and investigation

The analyst sorts the events by `TimeGenerated` to determine the order in which activities occurred. For example, repeated failed logins followed by a successful login may require investigation. If the same sequence is followed by account creation or an unexpected administrator-group modification, the activities should be correlated to determine whether an account was compromised.

- Affected account: Identify the account shown in the event records, such as `soc_user01`.
- Affected host: Identify the Windows VM from the `Computer` field, such as `Win-VM`.
- Activity sequence: Review authentication failures, successful login, privilege changes, and other related events.
- Potential risk: Unauthorized access, privilege escalation, persistence, or evidence removal.

Conclusion: Microsoft Sentinel helps SOC analysts correlate events, reconstruct an incident timeline, and identify potential threats. However, an individual event does not automatically prove an attack; the analyst must validate the evidence and surrounding context before recommending a response.

1d. Develop appropriate detection logic in Microsoft Sentinel using KQL Analytics Rules to identify suspicious activities generated during the attack simulations. Explain how these rules help detect potential security threats.

Microsoft Sentinel Analytics Rules automatically execute KQL queries at scheduled intervals to identify suspicious activities and generate alerts when the configured conditions are met.

1. Multiple Failed Logins

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count()
    by Account, IpAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 5
```

Purpose: Detects five or more failed login attempts for the same account and IP address within a five-minute window.

2. New Account Creation

```
SecurityEvent
| where EventID == 4720
```

Purpose: Detects newly created user accounts that may require investigation.

3. User Added to Administrators Group

```
SecurityEvent
| where EventID == 4732
| where TargetUserName =~ "Administrators"
```

Purpose: Identifies changes to local security group membership that may indicate unauthorized privilege escalation.

4. Security Log Cleared

```
SecurityEvent
| where EventID == 1102
```

Purpose: Detects the clearing of the Windows Security audit log, which may indicate an attempt to remove evidence.

Conclusion

These Analytics Rules help automate security monitoring, identify suspicious events, and generate alerts for SOC analysts. Analysts should investigate the related accounts, source IP addresses, affected host, and event timeline before classifying an activity as malicious.

1e. Document the investigation findings and recommend appropriate response actions based on the security events detected during the Microsoft Sentinel attack simulation.

The investigation was performed using Microsoft Sentinel to analyze Windows Security Events generated on the Azure Windows virtual machine. KQL queries were used to identify suspicious authentication attempts, account creation, privilege modifications, and other activities.

Investigation Findings

1. Failed Login Attempts: Event ID 4625 was analyzed to identify repeated authentication failures that could indicate brute-force activity.
2. Successful Login: Event ID 4624 was examined to determine whether successful authentication occurred after failed attempts.
3. Password-Spray Activity: Failed logins across multiple accounts were investigated to identify possible password spraying.
4. Account Creation: Event ID 4720 was checked to identify newly created accounts and determine who created them.
5. Privilege Modification: Event ID 4732 was analyzed to identify changes to local security group membership.

6. Process and Network Activity: Process execution, file creation, DNS queries, and network connections were investigated where the relevant telemetry was available.
7. Incident Timeline: Events were arranged chronologically to understand the sequence of activities and identify potentially related actions.

Recommended Response Actions

- Validate the alerts and investigate the affected account, host, source IP address, and event timestamps.
- If account compromise is suspected, disable or restrict the affected account and reset its credentials according to organizational procedures.
- Remove unauthorized administrator privileges and accounts after confirming that the changes are not legitimate.
- Isolate the affected host if evidence indicates an active compromise, following the organization's incident-response process.
- Preserve relevant logs and evidence for further investigation.
- Configure Microsoft Sentinel Analytics Rules to detect repeated failed logins, unauthorized account creation, privilege changes, and security log clearing.
- Monitor the system after remediation to verify that suspicious activity has stopped.

Conclusion: Microsoft Sentinel supports security monitoring, threat detection, and incident investigation by correlating events and generating alerts. The recommended response is to validate the evidence, contain confirmed threats, restore secure configurations, and continue monitoring. The simulated events demonstrate potential security risks; they do not independently prove that a real attack occurred.